

NEW EDITION
OF
HINTS
ON
Silvered-Glass Reflecting
TELESCOPES

MANUFACTURED BY
MR. G. CLAVER, F.R.A.S.

WITH DIRECTIONS FOR SILVERING, ADJUSTING, &c.

1877.

GEORGE CALVER,
HILL HOUSE, WIDFORD,
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HINTS

ON

Silvered-Glass Reflecting Telescopes.



f the various forms of Telescopes now in use, each has its own peculiar advantages; but the Silvered-Glass Reflector is undoubtedly gaining favour among our practical observers. A well-figured speculum, being perfectly free from chromatic aberration, gives, in a proper condition of the atmosphere, the finest possible definition of the Moon and planets, the markings and colours of these objects being excellently seen; while coloured stars, such as Albireo (β Cygni), or Almaach (γ Andromedæ), are exceedingly well shown, the beautiful contrast of the stars in the former being especially noticeable in a reflector. The advice of "F.R.A.S." (in the "English Mechanic," March 21st, 1873) as to the choice of a Telescope, may here appropriately be quoted. After expressing a preference for refractors when measuring close double-stars, he says, "But should the object of your correspondent be merely to regard the wonders and beauties of the Heavens, or notably, should he purpose to devote himself to the study of the physical structure of the Moon and planets, then by all means let him obtain the largest reflector he can afford; its absolute achromatism tells most astonishingly on these last-named objects." This is the opinion of one who has great practical knowledge of the different forms of Telescopes.

If Achromatic Telescopes of large aperture could be made as cheaply as reflectors, and in as convenient a form, they would doubtless be preferred for general star-work, although the aberrations, especially that of colour, cannot be so perfectly corrected. A silvered-glass reflector is, however, much cheaper than a refractor, and, when the aperture exceeds five or six inches, is much handier to work, and occupies less space, being only about half the length of an achromatic of the same aperture. It is true that a reflector will

give less light than an achromatic of equal aperture—but this is, in certain conditions of the atmosphere, a distinct advantage, the extra aperture to give the same light adding to the definition and penetrating power. An example of this is seen in the beautiful definition given by an unsilvered mirror on brilliant objects, as the Sun, Moon, and Venus. In large achromatics, the distressing excess of light has often to be reduced by diminishing the aperture or using a higher power than is convenient; and in such cases a lower and more suitable power can be employed with a reflector.

When the air is unsteady, the definition of Reflectors, owing to their tubes being open, is more liable to fluctuate than that of refractors, although when a reflector does not give good definition on account of the atmosphere, a refractor of the same aperture will certainly not perform satisfactorily. Sir John Herschel has shown that when the air in the tube of even an achromatic is disturbed in turning from one object to another, good definition does not return until it is brought to rest. In order to reduce the vibration of the air in the open tube, and also that of the stand, to a minimum, reflectors require to be very firmly and steadily mounted, and to have iron tubes.

Many of the specula made by me are now in observatories, where they have been compared with achromatics of first-rate quality with the most satisfactory results. For instance, several 6-1/2 inch specula were tried with two excellent achromatics of 4-1/2 and 6 inches aperture, when the planetary definition was considered to be superior with the reflectors; and the appearance of the star-discs, with equal apertures, differed little from the beauty and hardness of the images given by the achromatics. In viewing stars of great altitude, the use of the refractor is extremely inconvenient to the observer, whose position is necessarily very uncomfortable, while with the Newtonian reflector any part of the heavens may be observed with the same ease and comfort. In short, a good speculum, well mounted and situated, is sure to be both pleasing and satisfactory.

It is perhaps unnecessary to remind the reader that, when the defining powers of a telescope are put to the test, as much depends on the acuteness of the observer's eye and the practice he has had, as upon the perfection of the instrument and the fineness of air. It is a mistake to suppose that when the stars appear to be the brightest to the unaided eye, definition will be at its best, though this may happen; when it does so, the astronomer should make the most possible use of the opportunity, as such nights are very scarce. As a

rule, the best nights are those when there is the slightest possible mistiness of the atmosphere, and for the faintest stars absence of bright moonlight. The 5 inch mirror is guaranteed to divide stars one second apart with ease, and closer ones in very fine air. The 8-1/2 will split such difficult pairs as γ^2 Andromedæ, and μ^2 Boötis. An acute eye will master these stars with even a less aperture on very favourable occasions, γ^2 Andromedæ having been seen with a 7-inch stop of a 10-inch mirror, and Mr. Sadler, of Honiton Rectory, has split this star with his 6-1/2-inch telescope. It sometimes happens on favourable nights that the most difficult objects will be seen with the same telescope, which on other occasions had failed to show them as well as a much smaller aperture had done in very fine air. These remarks equally apply to the observation of minute stars and planetary detail. The amateur must follow the advice of the Rev. T. W. Webb, given in "Celestial Objects" (pages 15-17), and must cultivate that virtue applicable to all scientific investigation, namely, patience. The following interesting and difficult objects may be looked for, the powers used for their observation should vary from 150 to 300, and for the very closest stars up to 500, or even still higher.

TESTS FOR SPECULA.

LIGHT TESTS.

94 Ophiuchi. $5'' : 7, 13.$

58 Ceti. $3'' \cdot 5 : 6 \cdot 5, 14.$

γ Crateris. $3'' : 4, 14.$

15 Monocerotis. $25'', 15'' : 6, 9 \cdot 5, 15.$

τ Orionis. $15'', 20'' : 4, 15, 12.$

υ Ceti. $6'' : 4 \cdot 5, 15.$

ϵ Trianguli. $5'' : 5 \cdot 5, 15.$

179 Piscium. $3'' : 8 \cdot 5, 15.$

110 Herculis. $55'' : 5, 16.$

μ Andromedæ. $45'' : 4, 16.$

β Equulei. $35'', 3'', 50'' : 5, 13, 16, 14.$

85 Virginis. $30'' : 6, 16.$

55 Andromedæ. $25'' : 5 \cdot 5, 16.$

178^a Delphini. $20'' : 7 \cdot 5, 16.$

212 Libræ. $20'', 10'' : 6, 16, 8.$

14 Monocerotis. $10'' : 6, 16.$

94 Ceti. $5'' : 5.5, 16.$

δ Aquilæ. $1''.5 : 3.5, 16.$

SEPARATING TESTS.

33 Orionis. $2'' : 6, 8.$

52 Orionis. $1''.8 : 6, 8.$

δ Cygni. $1''.8 : 3.5, 9.$

2 Camelopardi. $1''.7 : 5.5, 7.5.$

π Aquilæ. $1''.7 : 6, 7.$

σ^2 Cancræ. $1''.4 : 5.5, 7.$

9 Leonis. $1''.2 : 7.5, 7.5.$

η Orionis. $1'' : 4, 5.$

ϵ Arietis. $1'' : 5, 6.5.$

4 Lyncis. $1'' : 6, 7.5.$

37 Pegasi. $0''.8 : 6, 7.5.$

749 ξ Tauri. $0''.8 : 7.1, 7.2.$

46 Arietis. $0''.8 : 8, 9.$

λ^a Cygni. $0''.7 : 5, 6.$

β Delphini. $0''.7 : 5, 5.5.$

20 Draconis. $0''.7 : 7, 7.5.$

287 Draconis. $0''.7 : 7, 8.$

178^b Delphini. $0''.6 : 4.5, 6.$

ϕ Draconis. $0''.6 : 4.5, 6.5.$

γ^2 Andromedæ. $0''.6 : 8.5, 9.$

μ^2 Boötis. $0''.5 : 8, 8.5.$

ι^a Boötis. $0''.5 : 4.5, 4.5.$

7 Tauri. $0''.5 : 6, 6.5.$

108 Draconis. $0''.5 : 9, 9.$

η Herculis. $0''.3 : 3, 8.$

42 Comæ Berenices. $0''.3 : 4.5, 5.$

ω Leonis. $0''.3 : 6.5, 7.5.$

Since writing the first edition of my catalogue, the writer has received many gratifying accounts of the success of the "Silvered-Glass Reflector." Many private letters have been received by him from observers, expressing their delight and satisfaction. This has been so encouraging that no pains have

been spared, nor any opportunities neglected, in turning to the best purpose every valuable lesson that continued practice and experience may have suggested in the manipulation of specula from time to time, in order to secure the best means for obtaining the highest excellence of defining power.

Many modifications and improvements have been made in working and testing mirrors, and special machinery and appliances constructed for large sizes. But to complete my conditions suitable for truly figuring and testing specula of considerable size, I found it necessary to remove from the traffic and tremor of a town to a still and tranquil situation in the country.

The truth of the curve is so sensibly and seriously affected by vibration constantly going on in and near a town, that it is liable to a variety of defects, and the surface becomes wavy and “plucked.”

As an instance, I may mention that my first and most convincing proof of the advantage of the stiller situation was tested by an 18-inch speculum (on which much labour had been bestowed); it was laid aside, but successfully finished after removal, and without undulations or any perceptible defect, and the Observer wrote me, that in good air, he “saw Sirius as a brilliant white dot, without a ray or appendage of any kind, and celestial photographs obtained with it are very fine.” Such results were exceedingly gratifying, as they were obtained with much less labour and uncertainly, and the tedious process of the final touches had not to be repeated so many times.

It is said that the celebrated Alvan Clarke, from the same effects of tremor, never finished an object-glass to his satisfaction above-ground; and Dr. Draper, testing his mirrors at the centre of curvature, to avoid draughts, &c. in an ordinary apartment, resorts to an underground one.

In this little book of “Hints” it may be useful to remind those who possess a speculum of fine quality that they are not produced by a “rule of thumb”—so to speak—and that the difference between a speculum and a really *fine* one, giving a maximum of defining and illuminating power, is the result of considerable labour and thought, and deserves careful usage.

The Rev. Cooper Key, an amateur of much experience, writes in the “English Mechanic,” that he was eight months (working sometimes eight hours a day), giving his 18 inch speculum its final touches and corrections.

A well known correspondent of the “English Mechanic,” “Hyperion,” tells us he found it impossible to test his 8-1/2 inch mirror in an ordinary room, and had to resort to a tunnel under the clay of his garden. Those who have the means and perseverance to make their own mirror, should be careful not to proceed with the finishing touches until an hour at night when their workshop, if in a town, is free from tremor.

First secure a well-ground and carefully-centred disc, let the polish be as perfect as possible before any attempt is made to figure. Care must be taken that every square or portion of the polisher is of the same consistency and temperature, that the disc may not be acted on irregularly.

To give the pitch this quality it must be well boiled and “pulled,” so that no air bubbles are in the squares, as these cause expansion or contraction as the temperature of the apartment varies, or that of the pitch and glass from friction.

It is much the best plan to keep the workshop to the same uniform temperature as the polisher was made for, allowing no draughts to pass over the mirror while working, nor the gas or lamp to be near. When the polisher is warmed—which it should be after laying aside for any time—it should be warmed *equally*. Neglect of the above in making the polisher, or any incautious handling of either the disc or polisher, will be sure to cause defects, which cannot be cured but by retracing the early steps in the fine grinding with an accurately centered tool.

The polishing successfully accomplished, the process of correcting for parallel rays is next proceeded with. At every step all possible care must be taken to prevent the mirror from running into an irregular curve. The importance of this cannot be too strongly urged, if a speculum capable of doing the *best* work is desired, as the curve *must* be true, regular, or uniform, to give the *highest* defining and illuminating power. An under-corrected mirror, if of a regular curve, will perform much better than a compound correction, exhibiting at the focal point much less lateral aberration.

A brief explanation of this may be of use as far as the limits of these pages will admit. It would be impossible to teach, by a mere description of methods, how to commence and finish off a speculum of good quality, even if every working secret and minute detail were unreservedly explained. The only way is to master sufficient theory, and the rest will come by prolonged

care and perseverance. I state this because amateurs who have been desirous of enjoying the gratification of observing with a mirror of their own making have written for information which they have been quite willing even to pay for liberally, but have felt disappointed, and perhaps thought it somewhat discourteous, on being told that what they wanted was impracticable and could not be satisfactorily attempted by the Optician in writing. In some cases I have finished amateurs' work on agreeable terms, and which has led to a pleasant correspondence or acquaintance. These remarks may prevent some future disappointment.

Every one is familiar with the fact that the parabola is the only concave surface that can reflect rays of light falling on its surface from an object at an infinite distance—such as the stars and planets practically are—to one common focus, or without aberration. This series or column of rays (which is equal in diameter to the opening of the speculum), reach the mirror without making an angle to it or to each other, they travel side by side, they are all of one length, and are reflected to a point, and are therefore all of the same length at the proper focal distance, viz., half the radius of the curvature of the concave. The properties of the parabola make the nearest possible approach to it of the utmost importance.

It has been said that a considerably under-corrected surface, if of a true and uniform curve, is better than one with less aberration, with zones or sections of various curves. To explain this, let us suppose an artificial star at some short distance, say 50 yards, the parabola would not form an image of this at the eye-piece without aberration; it has for this distance too short a focus for the central rays, and the best disc is inclined to the inner focus, because the rays from the object *diverge* towards it, instead of travelling parallel; and they reach the surface (the central rays compared with the marginal), at an angle equal to the semi-diameter of the mirror.

But, if instead of a parabolic mirror an elliptical one be used, which has one of its foci at 50 yards' distance, the image will be perfect. Now place the artificial star at 500 yards, the image will now be attended with perceptible aberration. The longer focus of the ellipse must be worked further and further from the mirror by shortening the focus of the central rays. Correct it for this distance, and again remove the artificial star to a still greater distance, repeating the corrections as before and carrying the outer focus towards the object, and the inner towards the mirror, until the rays from the

object become more and more parallel, and the curve is nearer and nearer to the parabola, or that eccentricity of ellipse which acts better and better for distant objects or parallel rays.

It is evident from this that if the ellipse is corrected very considerably towards the parabola, without irregularities, and *every part* of the surface corrected regularly from the centre to the edge (no part hastening more than another), that such a surface, though under-corrected, is much better than if one portion is fully and another under-corrected, especially if the more imperfect portion is towards the edge. Such a compound correction may show little or no *outstanding* aberration at the focal point, but the rays do not find a focus at the same regular pace as they would from a regular curve—one edge of a zone will be coming into focus when another would be going out. With the focussing screw they are focussed to the place where they appear to collapse, and are most satisfactory, though, *in reality*, the rays at the best place bend over each other, and the definition is imperfect. From a star, which is only a point, this may be more tolerated, but on the planets, where the image has a sensible diameter, and is perhaps magnified many times, this imperfect curve is very inferior to the regular one, whose error is all of *one kind*. The light is all there *somewhere*, but not with any good effect. There is no *proper* illumination or definition, as rays are employed which are crossing the optical axis at *varying* angles, and the result is confusion.

So the amateur who sets himself the pleasant task of making his own speculum (for there are many who can better afford the labour than the capital to purchase, and whose capabilities are thus superior to their means), need not be discouraged and give up the pursuit because he cannot obtain the best results by getting a perfectly parabolic glass. But if he has been successful in removing a considerable amount of the spherical error, and advanced to the elongated ellipse by maintaining a truthful curve, “let well alone” with this disc, and proceed no further, but commence *another*, taking care not to alter the first until the second is *better*, and *then* an improvement of the first may be attempted.

In the second attempt, should the amateur lose control over the *regularity* of the surface, let him try it as an experiment against the first on the planets, and he will not fail to appreciate the difference, and will be stimulated by

fresh courage to get as near to the parabola as possible with the same accurate curve.

To produce a true and uniform curve is, however, the acme of troubles, whether it is desired to obtain the spherical, or parabola, or any other curve. It is generally supposed that the spherical curve is a very easy matter, so easy indeed that it is difficult to avoid. This is a very great mistake—a spherical curve of undeviating truth is as difficult a problem as a true parabola. The spherical curve is the *only uniform* curve, it has but *one* focus, and the polisher must coincide with, and be of the same radius as the glass, at every instant. This is why the optician strives to obtain a semi-polish with the grinder to lessen the risk of losing his curve on the polisher, for the curves of an object glass are spherical. The curve most liable to be obtained by the amateur is the spheroidal, a curve with its marginal rays *shorter* than the central, or half the radius of curvature.

There are no means with the *telescope* of telling the spheroidal, approximating to the sphere, *from* the sphere. There are *no* means of analyzing the exact character of a curve *equal* to certain methods at the centre of the curvature, but to accomplish this requires much practice and observation, with “surroundings” perfectly free from vibration. If the amateur can overcome this, and lives in or very near a town, he should only work at the polishing and figuring during the late hours of the night, when traffic, &c. have subsided. Then, by carefully preventing any draughts in the apartment, and with the mirror of the same temperature as the air in the room, he will then be able to see how varied and numerous are the chances of error in working a mirror, and the great care necessary to avoid or cure them. He will find the surface exceedingly prone to receive zones and irregularities during work, and much more so than to “work true.”

The necessity for avoiding incautious handling or heating may be realized by the following little experiment when one can manage and understand it:—Place the tip of one finger on the surface, as it hangs in the dark room ready for testing, and with very gentle pressure let it remain long enough to spell one’s name; it will then be seen that the feeble heat of the finger has, by expansion, raised a mound on the surface of the glass, and though this amount of swelling must be *very* small, yet it is enough to cast a shadow across the surface, as if something were laid on it, and quite ten minutes will elapse before the heat will leave this spot and the surface again become

level. Now if the polisher were placed on the glass while this hillock was there, a permanent hollow would be the result. For a full account of these methods (of which Foucault was the discoverer) the reader is referred to Sir John Herschel's and Dr. Draper's works on the telescope.

Care must be taken not to leave too much aberration, as then the central disc is formed too positively outside the focus, and the rest of the light from the object appears as obtrusive rings and false light. The over-correction is bad, and acting as a negative lens the disc is formed too near the mirror. Such a correction, besides being objectionable on almost all classes of objects, prevents the use of the "Barlow" lens, and acts badly with all kinds of positive or Ramsden eye-pieces, which improve *under*-corrected, but "make bad worse" with over-corrected surfaces.

If these few and brief hints should stimulate the industrious and persevering student to make his own telescope, and thus enjoy the fruits of his own labours (which may add a relish and a pleasure to his astronomical work), they will have served the purpose for which they are written.

After a "Hint" on the choice of focus for the mirror, it only remains to say a few words about the plane, as this, with the large speculum, are the only parts that need be home-made as far as the optical work is concerned. Never adopt a "dumpy" for general use where high powers are sometimes wanted, for small and moderate sizes, say 8 to 12 diameters, and for large sizes not less than about 6 diameters of the mirror, as the larger ones practically admit of a shorter focus. The short ones can be mounted somewhat cheaper, but never choose them on *this* account, they will not make so satisfactory an instrument, and no adaptation of "Barlow's" will make it so.

To make the plane, provide three well-turned metal discs of 7 inches or 9 inches diameter. Iron is the best, as the work will go down closer, and the plates or discs may be very thick—say an inch—and not so liable to "spring." These turned discs must be scraped and ground perfectly flat on each other, until they are in good contact all over, so that there is nothing between the faces when testing them. In the earlier stage of "truing," oil and colour can be used. When these discs are proved true, a disc of plate glass, same size, is cemented with pitch on one of them, or on a thick disc of glass, and care must be taken that it is not strained while on the block during working. Truly grind this plate on one of the tools to a fine semi-polish. Polish on the same tool with a piece of thick silk or alpaca, laid over and

cemented down by a solution of resin dissolved in turpentine (as much resin as the turpentine will dissolve), then, with the tip of the finger, thinly smear over the tool and bind round the edge with cord, and the silk will keep in place. Fill up the texture of the silk or alpaca evenly with damp rouge, keep it uniformly damp, and never let the rouge and water work about. It may be polished on very hard strained pitch, but pitch for the amateur is not so safe, as it is liable in his hands to alter its form and destroy the truth of the plane, but it is the quickest and handiest if it can be managed.

After the polish is *perfect* remove the plate from the support and cut into squares a little larger than the size of the intended flat. Test these in the telescope on a *star*. If one plane turns out bad, the whole will most likely be so, and another plate must be worked with renewed energy and care, for a bad plane will spoil the action of the speculum however good, for there is no way of counteracting the curved surface of the flat or second reflector. The edge of the plate for about an inch should not be used.

When a good plane is found it can be edged by turning a piece of wood a little less in diameter than the minor axis of the plane required, turn the end square, and mark a line around it distant from the end equal to the diameter, and cut through to the opposite edge, and it will give the oval and will appear round at 45°, this will mount nicely in brass tubes of the proper size, and a cover should be made to fit easy.

“THE EQUATORIAL.”

Fig. 1 is a modification of the German principle, and it should be a sufficient recommendation to remark that it is the principle used by a maker of such experience as Mr. John Browning, and is, without doubt, the very best style of mounting for a fixed equatorial, especially when clockwork is employed, because clock power is applied to the polar axis itself direct from the driving worm.

The driving part, viz., the worm and wheel, which is out of the observer's way (being between the standards), is not liable to meet with accident, and the driving wheel being near the lower end of the axis, is at the most rigid part, viz., the foundation of the instrument. As it has a long and very firm

polar axis, and is connected with the foundation plate, it secures the utmost steadiness and freedom from strain.

An equatorial mounting, with two long and stiff shafts for the axes, has always the advantage of firmness; the *holding* portions of the instrument being in masses, are not liable to receive injury from blows, and thus be put radically out of order. *No means should be resorted to to make an equatorial of light weight*; an instrument cannot be light and slim and at the same time *firm* and *steady*—no amount of steadying rods, splines or strings, will make a slender tripod for a refractor so steady as a firmly made one, with proper size and weight in the parts.

Another important advantage of this kind of mount for a large size is, that although it has considerable weight, it is very convenient to move and set up, being built of convenient parts, which are easy of separation and removal. The uprights or standards are separate, and are bolted to the bed-plate, the upper and lower discs are readily detached from the stand. The cradle and tube are in this construction separable. The tube of the telescope being suspended over the *side* of the stand, is in the most convenient position for observing objects at any altitude, as the stand is out of the way and clear of the telescope. The various disadvantages and objections to the driving clock being *carried* by the telescope are here avoided. The clock, which is large and very powerful, is bolted down to the iron bed-plate, and the telescope, not having to carry the clock or weight, the balance is never altered nor the rate of the clock disturbed, and thus a strong clock, keeping regular time, and working much longer can be used. They will drive 2-1/2 hours. It will be readily understood that a driving clock to enable the instrument to do the most exact and best work must be powerful and a good time-keeper. Lord Lindsay has said that “the clock should have twice as much power as is *used*,” otherwise the spindles, &c. are pent up, and it is moreover sensible to any extra weight or work being put upon the instrument.

To give an idea of the efficiency and regularity of this clock, I may mention that a Newtonian telescope of 18 inches aperture, intended to carry a photographic camera of 112 lbs. weight, needed no additional provisions to do the extra work, the rate and power not being affected. There is a mechanism in the clock “for making up of time” and in setting the hour circle can be used with a joint handle or hooks as a fine screw-motion, and can be applied whether the clock is going or not. There is also a provision at

the foundation for throwing the instrument out of the meridian, to follow the motion of the moon or planets with the clock and “maker up.” The above mounting is equally suitable for “Cassegrains” and “Newtonians,” to both of which forms it has been successfully applied up to 18 inches of aperture. It is well suited for large telescopes.

Fig. 2 is a mounting on the same principle, but with a shorter polar axis, and the column is in one casting. It is well suited for moderate sizes, and the circles, &c. are applied exactly as in Fig. 1, but when a clock is wanted there is no mounting equal to the former.

The “Educational” is a plain 6-1/2 inch, of 4 to 6 feet focus, and is made to this pattern, with revolving body and screw-motion for following. It is made as portable as it can be, and is a steady, good-working instrument, and much approved of.

THE UNIVERSAL ALT-AZIMUTH.

(FIG. 3.)

In this mounting I have seen little or no alteration needful, except that the tube is now made to balance so that the eye-tube can be reached for objects in the zenith without the observer having to elevate himself, and the elevating rod can be clamped without the lever, the legs of the stand are more curved and have more spread in them.

This “Alt-Azimuth” stand has met with much approval; and where portability is of consideration and the observer has to set up and remove the entire instrument after every night’s work, this, or the Angle-Block stand, will be found the most convenient, more so than an equatorial in any form. There is much less weight to remove, and being in three convenient parts is more readily put together and separated in the dark. If the stand and trunnions can be left in the open air a very small covering will serve to protect them, and then the tube alone has to be removed. The equatorial cannot be too highly recommended where it can be a fixture and undisturbed, as when once got into proper adjustments its advantages can then be realised, but not unless it is a permanent fixture. The circles of a portable equatorial can only be used for very rough reading, and

consequently the adjustments are never in order, and the readings are of very limited use indeed.

THE IMPROVED “ANGLE-BLOCK” STAND.

(FIG. 4.)

With respect to the tube, it is mounted like the Alt-Azimuth, but with the plane of the horizontal movement corresponding to the latitude of the place of observation, and therefore following objects with one screw movement is in reality a telescope with equatorial motions. It need not be of heavier construction than the Alt-Azimuth, and there is not the double weight of counterpoising, &c. as in the equatorial, and when circles are not desired this will be the most economical, handiest, and easily managed instrument, as it partakes of the equatorial form or motions. The telescope tube is well balanced, and the declination movement is easy and free, and fitted with a clamp screw, so that when the instrument is turned on an object that object can then be followed by an endless screw.

It can be made to suit any latitude, and by the addition of foot screws on a level floor, can be got into suitable adjustments, and if it has to be removed (for the stand is very little heavier than an ordinary Alt-Azimuth), “guides” can be provided, so that it can go very approximately into the exact place again when brought out for use. They are made, when desired, with a revolving body and fine screw-motion in declination.

THE “POPULAR REFLECTOR.”

WITH ANGLE-BLOCK STAND.—(FIG. 4.)

The speculum is 5-1/4 inch diameter, and carefully figured, and is recommended as a very useful instrument. The size and power of this telescope has been adopted as that most likely to meet the means and requirements of a large number of amateurs. Many prefer to commence astronomical observations with “*something inexpensive*,” and are led to begin with the popular and well advertised 3 inch refractor at £5. These, except in a few chance cases, are sure to prove disappointing, and are perhaps the cause of their giving up any further attempts to follow up the

subject of astronomy, which may otherwise be so pleasantly and profitably pursued with a reliable instrument.

It is well known that a less aperture than 4-1/2 inches is insufficient to give the observer a satisfactory idea of the varied and most interesting details of the planets, and the ever-changing outline and tone of the belts and markings of Jupiter, Saturn, and Mars. Large apertures bring out *details* when the smaller ones can only show a *general outline*. The former also delineates more with a lower power, providing the focus is of proper length.

The planets and nebulae cannot be seen to advantage without aperture and focal length. The field of view is then flat, the object is properly illuminated, with sharp and crisp definition, and is also *much* less subject to annoyance from tremour, through the necessity for constant adjustments in the fields, &c., as is the case with small apertures, for the object is magnified more in proportion. A certain magnifying power is necessary in order that the object may be sufficiently large to scrutinize; and this, whether the aperture is large or small, must be from about 150 to about 300 times for the planets. But 150 on a 3 inch is a high power for the quantity of light obtained, to say nothing of the separate consideration and advantage of long focus. Aperture is a quality or function of the telescope considered separately from light or focus. For instance, suppose a 5 inch speculum is so thinly silvered that it gives the exact degree of light as a 2-1/2 inch refractor, the defining and separating power of the 5 inch aperture would be very superior to the 2-1/2 inch.

The “Popular Reflector,” with 5-1/4 inch speculum of 5, 5-1/2, to 6 feet focus, will be found very suitable, and if its illuminating power is not greater than a 4-1/2 inch refractor, its defining and dividing powers are superior. By choosing the above focal lengths, according to circumstances, the observer can reach the eye-tube, while standing erect for objects in the zenith, and it is a great mistake to suppose that the shorter the tube the more handy it becomes. These foci will not require a “Barlow” lens to flatten the field. The “Barlow” is very useful for short foci when the aperture is considerable, as it improves the imperfect correction for spherical error, but this is much better corrected in the mirror itself than by a “Barlow” lens, which cannot be used without more than one disadvantage. There must, by its insertion, be some loss of light, which can ill be spared with small instruments, and when used to obtain magnifying power there is some disturbance of colour, and this subtracts from the beauty and purity of the definition of a reflector. There is

nothing equal to a good eye-piece to obtain *power*, and *flatness* of field by the *curve* of the speculum.

With the “Popular Reflector” and an outlay of a few shillings on some popular books, such as the excellent work “Celestial Objects,” by the Rev. T. W. Webb, and some first-class publications by Mr. R. A. Proctor, especially his smaller star atlas, &c., the amateur can compare the work he is then capable of doing with a large and expensive refractor which might be beyond his reach.

THE ADJUSTMENTS OF THE EQUATORIAL.

When the inexperienced amateur purchases an equatorial with circles, he should not be without the third edition of “Chamber’s Handbook of Descriptive Astronomy,” which, besides being an excellent book in other respects, is a really practical guide to the use and application of the Equatorial, and is indispensable to the beginner. He will there find the fullest details of the adjustments to any degree of exactness. Besides many other matters he will be instructed in the use and application of apparatus to the perfect Equatorial, including all kinds of eye-pieces, micrometers, &c., &c., as well as other optical instruments and accessories.

The Equatorials described in this catalogue are provided with every means of adjustment. The cradle contains powerful screws to set the line of sight at right angles to the declination axis, and shifting screws to place the polar axis in the meridian and to the correct elevation, and with care and a few experiments with the adjustments, and by observations of some catalogue stars, the various adjustments will soon be correctly made, and the verniers set accordingly.

THE CASSEGRAIN.

This is a form of reflecting telescope but little known. This is rather strange, since it is a very much better principle than the “Gregorian,” so well known to the old observers. Herschel says it admits of a theoretically perfect telescope. Compared with the “Newtonian” it has its advantages and disadvantages. Its principal advantages are, first, the shortening of the tube,

which in large telescopes is sometimes very important. Secondly, the observer has not to *ascend* to the eye-tube, the observations being made at the lower end, as with a refractor. The “Cassegrain” has a flatter field of view, owing to the action of the curve of the second reflector causing the rays to travel twice the distance, and, adding the element of magnifying power, the eye-pieces need not be composed of small lenses. The adjustments are perhaps a little more trouble, as the line of collimation must be carefully attended to, this requiring only a little more care can soon be accomplished, and then the definition of a good “Cassegrain” is very pleasing.

Amongst its disadvantages is the necessity for the observer to gaze upwards as with refractors, which, when the object is at a considerable altitude, is distressing, this is one of the reasons why the “Newtonian” is so pleasant to use, on account of the natural and easy position of the observer. The eye-piece being a fixture, it is not quite so convenient to use some of the accessories of the telescope. But there are means to overcome these drawbacks, and so make the “Cassegrain” even more handy than the “Newtonian.”

I have mounted an 18 inch speculum of 12 feet focus in the “Cassegrain” form, so that objects at any altitude could be observed with the greatest possible ease. A plane was fixed near the large mirror to receive the rays from the convex reflector and to throw them out to the side, illuminating apparatus were fixed here, and micrometers, &c. used, as if it were a “Newtonian”; the tube was thus made shorter, and the flat field of a long focus realised, but there would be a little loss of light in consequence of an extra reflection, but this, with a large aperture (and the fact that the “Cassegrain” gives a little more light than the “Newtonian”) can better be spared, considering the convenience gained. The observer is not elevated for any altitudes, and a large telescope is actually occupying less room than a small one. It can be used with or without the diagonal.

I have, by request, fitted the “Cassegrain” with means for two observers to view the same object at the same time. A perforated plane was arranged to receive a portion of the converging rays and throw them to the side of the tube into an eye-tube, and the remainder passing on to the eye-tube at the proper place, two images are thus formed, and can be magnified at will and

viewed simultaneously. The perforated plate was so arranged that it could be removed at pleasure.

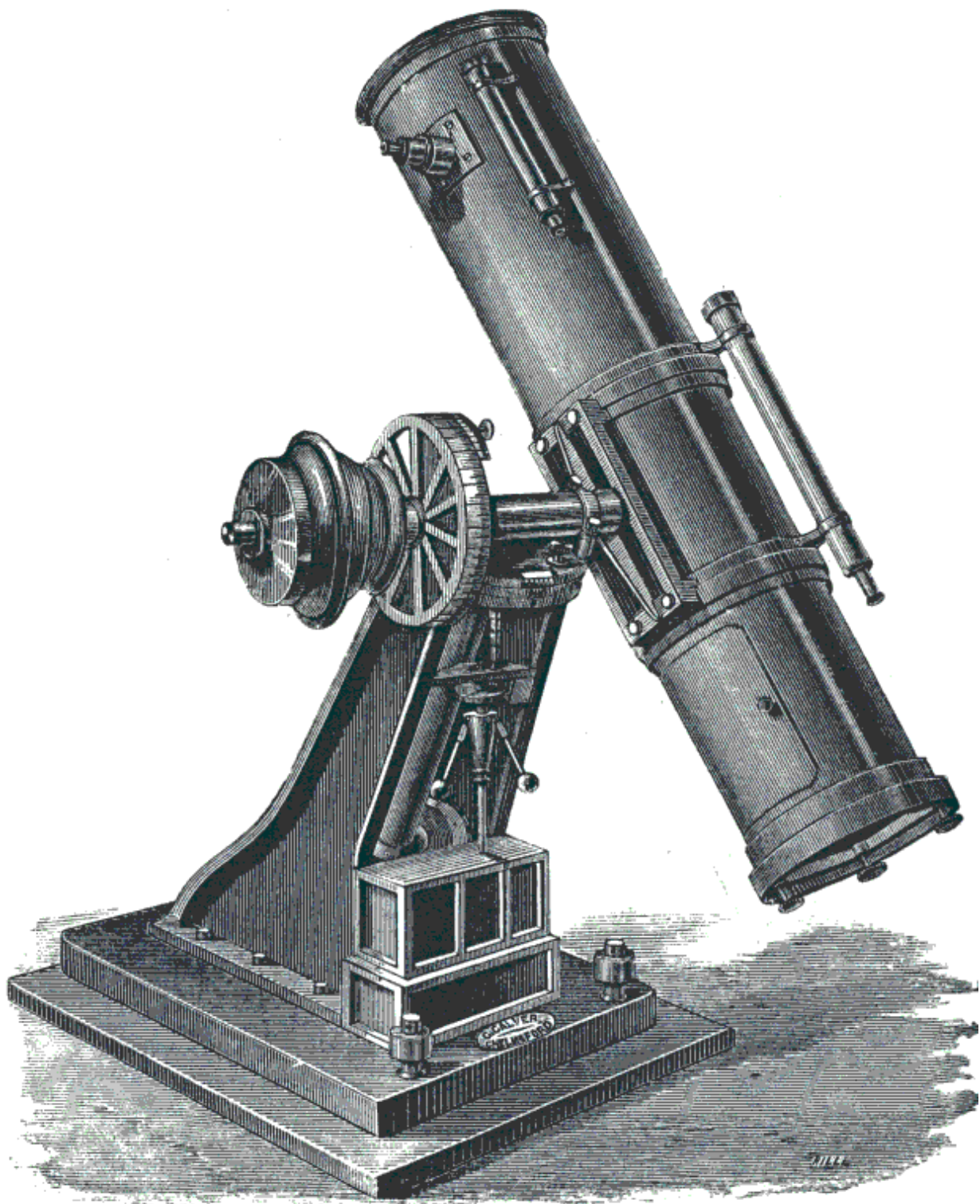


FIG. 1.

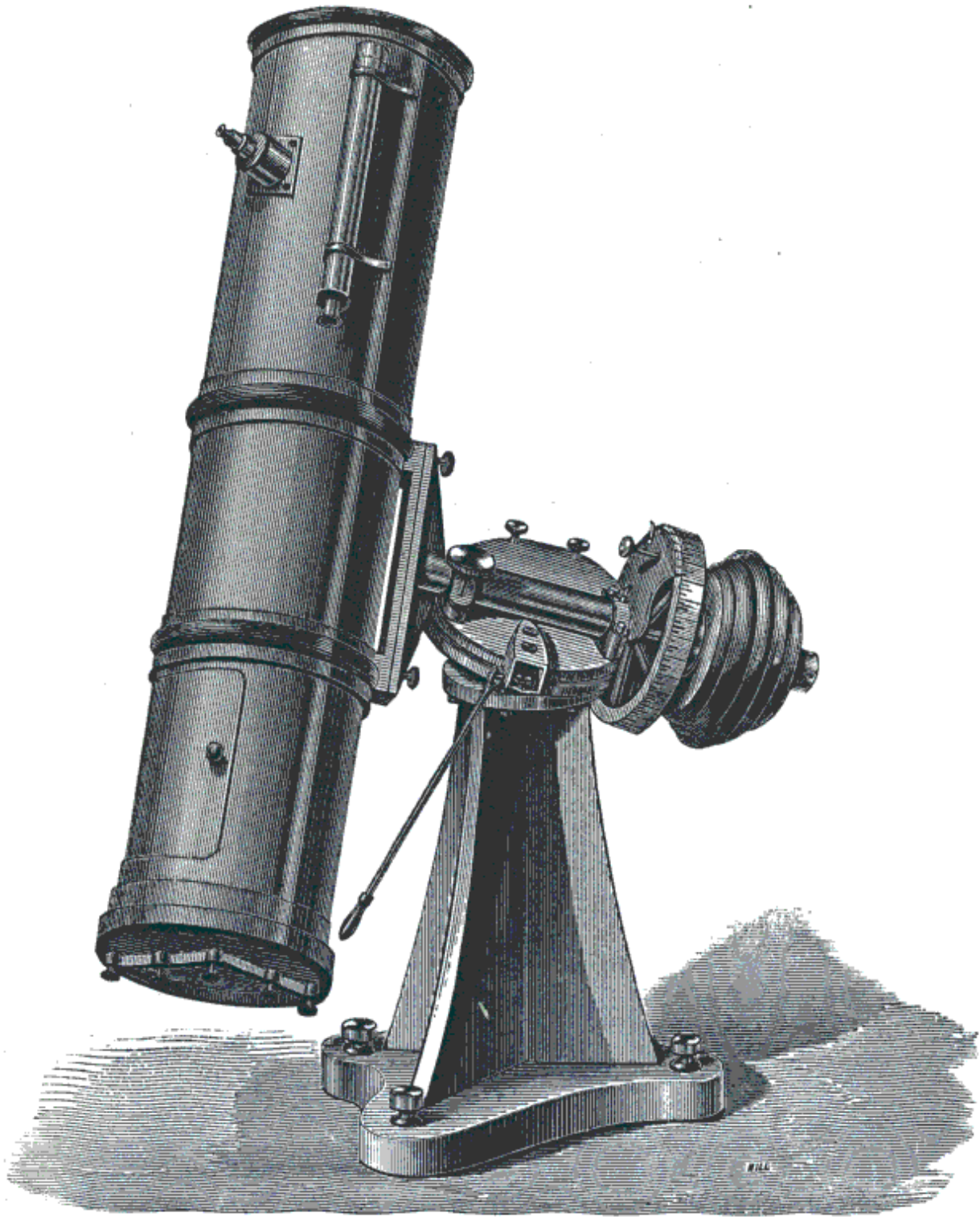


FIG. 2.

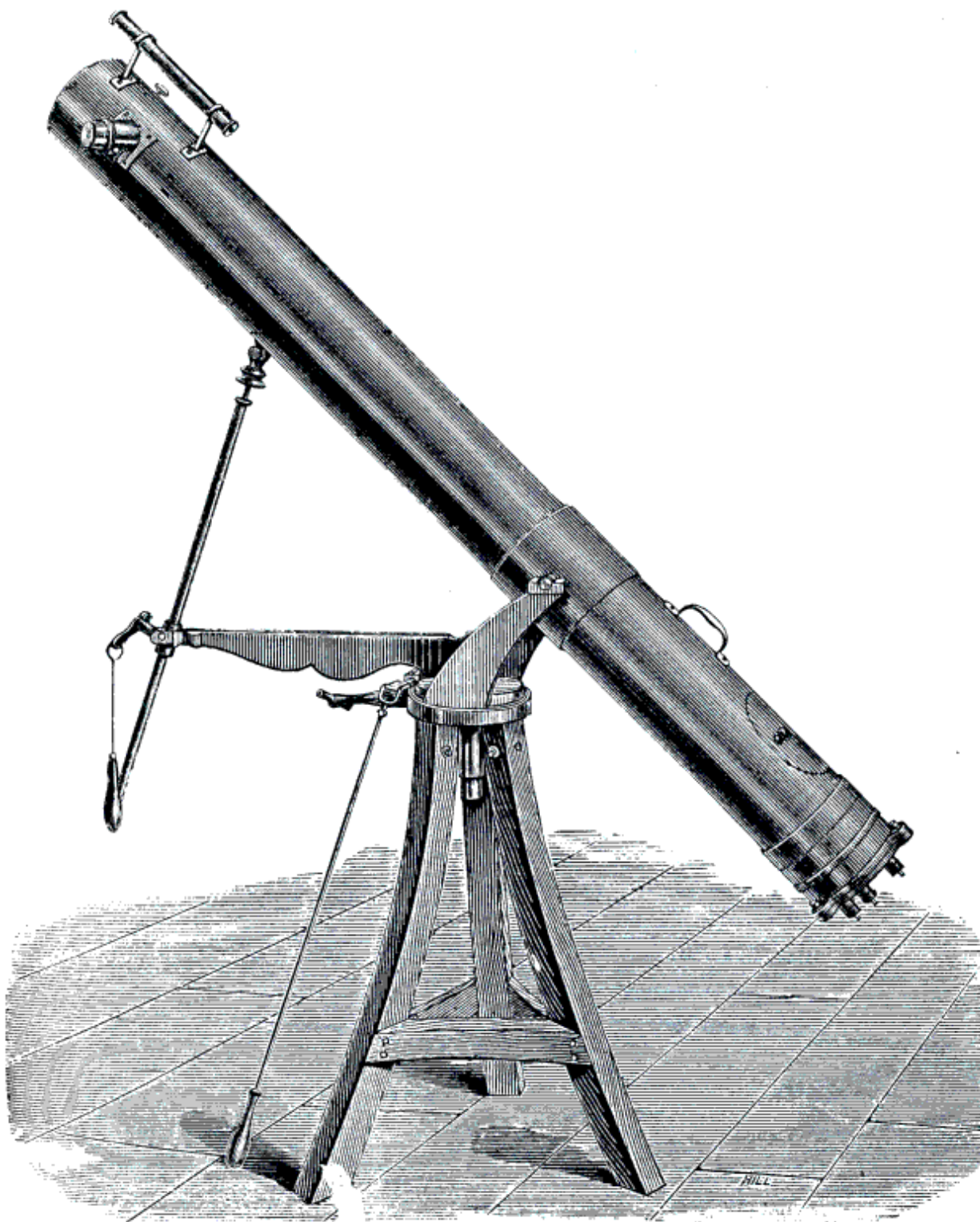


FIG. 3.

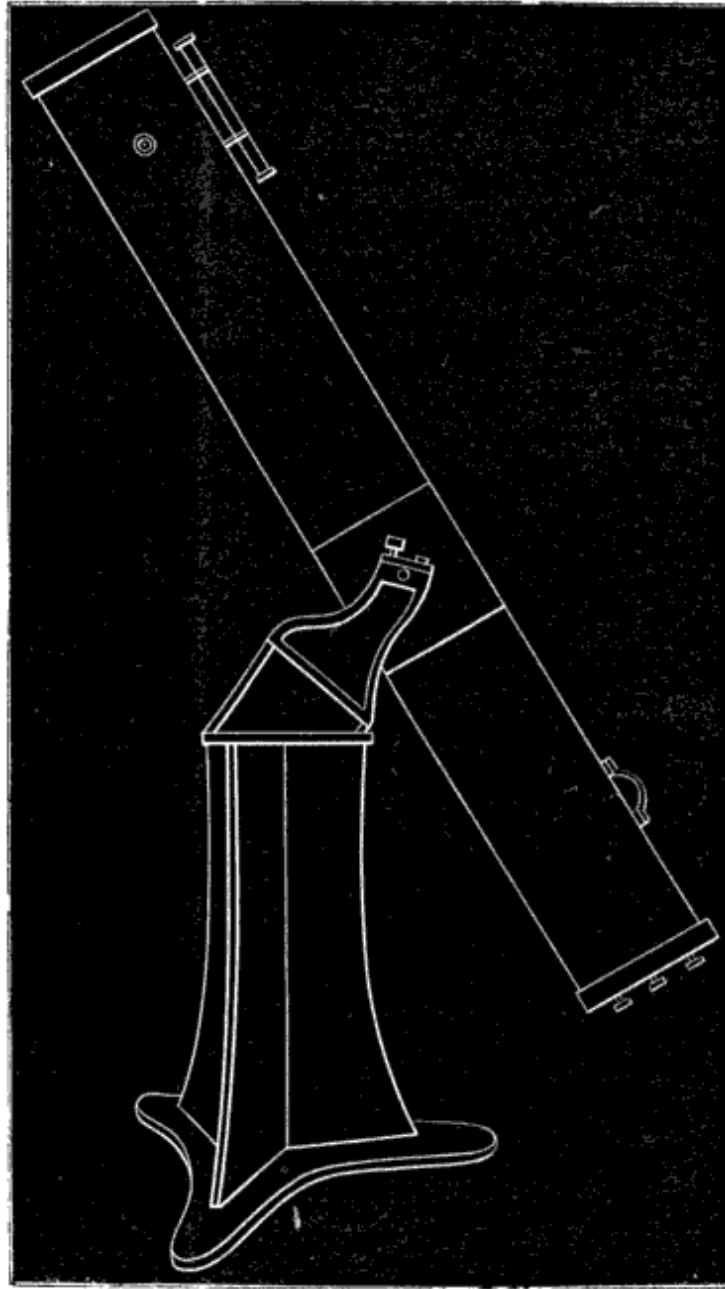


FIG. 4.

THE ALT-AZIMUTH STAND.

Fig. 3.

The Telescope, being balanced on trunnions, can be moved from an elevation approaching the zenith to an almost horizontal position. In order

that it may be secured anywhere between these extremes, attached to the upper part of the telescope is an iron rod, which, sliding through the end of the arm of the stand, can be there clamped. The telescope will now be clamped in *Altitude*. As the progression of celestial objects will apparently be very slow, resource must be had to the smoothness of motion obtained by a screw. The upper end of the Altitude rod is therefore tapped to receive a long screw with a large milled head, jointed to the telescope body; by revolving this head the telescope is raised or depressed accordingly as the screw is unscrewed, or the reverse. It is necessary that the screw should be withdrawn some way from the rod before clamping it, preparatory to following an object which has passed the meridian, or is setting; as perhaps, just when the clearest vision is obtained, the observer may be annoyed by the screw action being suddenly stopped by the milled head coming in contact with the top of the rod.

When viewing objects near the zenith, and the focus long, the handle attached to the clamp will be found useful, as it can thereby be reached without leaving the finder. The handle may be so placed that a downward push should clamp, and an upward pull release.

The second motion in the Alt-azimuth Stand, namely, *Azimuth*, is obtained as follows:—The strong iron disc which forms the upper fitting of the legs has its surface accurately turned. On this revolves an iron disc, rather less in diameter, to which the trunnions which support the telescope are attached. The main axis of this disc passes through the centre of the lower disc, and then through a hollow bearing tube, a continuation of it. All these fittings having been most carefully turned and ground together, great steadiness, combined with facility of horizontal movement, is ensured. In order that this motion can be communicated as evenly as possible, resource must be again had to a screw which is thus applied. Just within the circumference of the lower disc is a narrow groove, turned to such a depth that the ring which is thus separated from the main disc is still firmly held to it by the uncut portion. An iron clamp grooved to this ring holds the nut of a long screw, the plain end of which is jointed to the upper disc. When this clamp is fixed to the ring, any motion given to the screw will act on the upper disc, and cause it to revolve, and thus the whole telescope will be slowly moved in *Azimuth*. The advantages of this plan are many, the most important being the rapidity and ease with which the telescope can be shifted from one object to another, even to those in contrary directions; all that is necessary being to release the

clamp and turn the telescope to the object required. The clamp being carried round with the upper disc, can be fixed directly the desired position is obtained, when the screw is at once in action. Should it happen, whilst following an object, that the screw becomes exhausted from the joint and clamp coming together, the clamp should be released, and the screw turned sufficiently in the reverse way to bring it into action, the weight on the upper disc keeping the telescope meanwhile in position. If this operation be rapidly performed, the whole length of screw can be brought in play before the object has left the field of view of the finder, and can thus be easily refound with the higher power of the telescope. Motion is applied to the screw by means of a Hook's joint, named thus from its inventor. This joint being furnished with a long handle, enables the observer, by means of it, to move the Telescope in *Azimuth* at any rate, and without removing his eye from the eye-piece.

It will be seen from the preceding remarks that by means of the vertical and horizontal screw motions, the telescope, when clamped, can be moved in any direction with the greatest facility, permitting a celestial object to be observed with high powers for a considerable time, and with the greatest pleasure and comfort to the observer.



ADJUSTMENTS.

These instruments are always sent out in correct adjustment, and with moderate care during transit, and afterwards, will remain so, but as the performance of the instrument greatly depends on the accuracy of the adjustments the following instructions will enable the observer not only to ascertain whether they are perfect, but also to render them so if found defective. These adjustments are by no means difficult, and will be easily understood by attention to the following remarks:—

Into the draw tube screw the “adjusting piece,” which is a small brass circle with a hole in its centre about $\frac{1}{20}$ of an inch in diameter. (The draw tube should be in about the same position as when at focus with an eyepiece.) Place the large mirror in its cell in the tube or body of the telescope, taking care that the three bayonet-joint pins are correctly placed, that is, with grooves pointing downward. They will be found to drop easily into their corresponding holes; care must however be taken that the grooves have gone well home. Both the speculum and small mirror, or “flat,” must be uncovered. On looking through the aperture of the adjusting piece, if the mirrors are in correct adjustment their reflections will be seen as follows:—the small oval mirror being placed at an angle of 45° will appear circular, and reflected exactly in the centre of this circle will be seen the bright image of the large mirror with a dark round spot in its centre, as shown by Fig. 1. This dark spot is the double reflection of the “flat,” and should be concentric with both the bright reflection of the large mirror and the circular outline of the “flat.” All these should also be perfectly concentric with the circle given by the stop in the draw tube. Should these circles not be all central the adjustments are not perfect and must be rectified as follows.

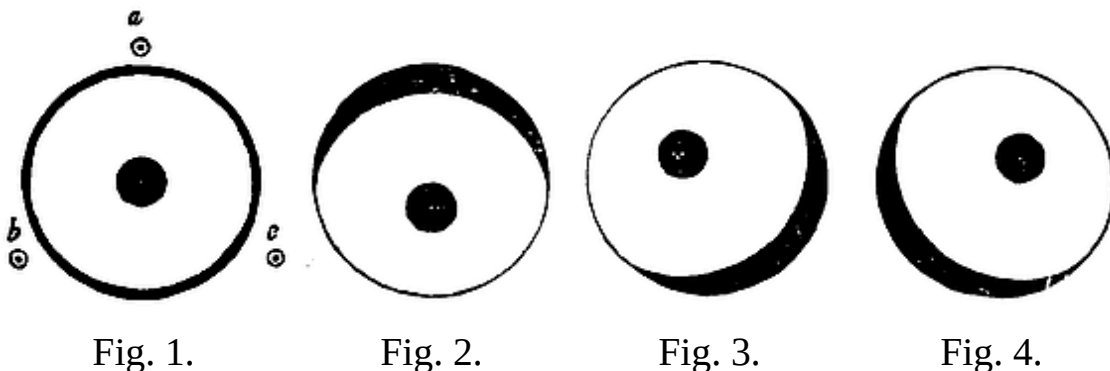
To adjust the “Flat” or small diagonal Mirror.

If the bright reflection of the large mirror is seen as a perfect circle, but not exactly in the centre of the “flat,” the latter requires adjustment; for this purpose loosen the milled head screw at the middle of the back of the “flat”

which in large instruments is made sufficiently heavy to act as a counterpoise to prevent vibration. This will allow the “flat” to be rotated by the hand vertically with respect to the tube of the telescope. Bring the bright circular reflection of the mirror exactly central in this direction, and fasten in position by screwing up the milled head screw or counterpoise. If the circular reflection is quite central no further adjustment is required, but if not, then, after completing the vertical adjustment, as described, proceed to make the horizontal adjustment by turning in one or other direction the milled head of the horizontal adjustment screw, situated in front of the vertical adjustment screw; this will bring the circle of light exactly into its proper horizontal and central position, and the adjustments are then completed.

If the bright reflection of the large mirror is *not* seen as a perfect circle, and the small dark spot not in the centre, the speculum is out of adjustment, and consequently the adjustment of the “flat” is best performed by removing the large mirror or speculum with its cell and so arranging the body of the telescope that on looking through the “adjusting piece” a large sheet of white paper spread on the ground a short distance from the open end appears as a white circle of light reflected in the “flat.” Now bring this white circle exactly into the centre of the flat precisely in the way described above, and on this being accomplished replace the speculum uncovered with its cell, and proceed.

To adjust the large Mirror or Speculum.



When in perfect adjustment the large mirror viewed through the “adjusting piece” should appear as before stated, as a complete bright circle, with the

image of the “flat” as a smaller dark circular spot exactly in the centre (Fig. 1). Should this not be the case it must be rectified by means of the three adjusting screws at the back of the cell. Proceed as follows:—First unloose (by a few turns) the small clamping screws which pass through the larger hollow adjusting ones; then, on looking through the adjusting piece the relative position of the image of the “flat” should be carefully noted. If the dark spot is nearer the bottom, and consequently more of the top part of the bright circle is seen (Fig. 2), the mirror reflects too much of the upper part of the tube, and therefore the top of the mirror leans too far back and must be pushed more forward by screwing in the top adjusting screw (*a* Fig. 1) a little at a time till the dark spot is central. Should the spot be towards the top (Fig. 2 *inverted*), the reverse holds good and the top adjusting screw (*a*) must be *unscrewed* or the two other adjusting screws turned in. Should the spot be towards the left (Fig. 3), screw up the right adjusting screw (*c* Fig. 1). If towards the right (Fig. 4), screw up the left adjusting screw (*b* Fig. 1). The dark spot is always furthest away from that part of the tube which is too much reflected, and from the adjusting screw that must be turned in to correct it.

When all the adjustments are considered perfect, as in Fig. 1, the small screws are to be clamped up to keep the adjusting screws in position. If, after the greatest care has been taken in adjustment, a flare should appear on looking at a star (say of the second magnitude) with an eye-piece of a high power, and the diffraction rings are not quite concentric, it can generally be rectified by turning the large screws of the mirror round in cell a little at a time. If this does not remove the flare the adjustment of the “flat”. is not sufficiently correct and must be altered by means of the screws at its back. If the flare is at the top or bottom of the star the “flat” must be very slightly revolved by the hand, after unloosing the middle screw, and when correct reclamping it. If at either side, namely, in the direction of the major axis of the “flat” and in a line with the tube, the “flat” must be altered by the long screw. It is always advisable to leave the telescope for a short time undisturbed, especially if, on first looking at a star, a flare should appear, as these appendages often vanish when the instrument has been for a short time in the air. These adjustments may at first appear somewhat difficult, but are rendered remarkably easy by observing how the different screw movements alter the positions of the reflections. A useful test as to the correctness of the adjustments may be obtained by viewing a star with a high power eye-piece

out of focus. When in the centre of the field it should appear as a bright luminous circle with a circular dark spot in the centre, the size of the bright circle diminishing as the focus is approached and the dark spot remaining central.

The cell mount of the large mirror can be removed from the tube and replaced without disturbing its adjustments, but it is very advantageous if the entire instrument can always be left undisturbed when not in use in an observatory of light construction, having a skeleton revolving dome, covered with well oiled canvas or calico, and made with a wide opening and large shutters. The Rev. E. L. Berthon has described in the "English Mechanic," October 13th, 1871, an observatory of this kind most admirably suited to shelter a reflector, as the temperature inside would be as nearly as possible that of the external air, and no annoyance from damp would be experienced. Where an observatory is not practicable the telescope might be protected by a close fitting covering of like make. Both the large and small mirrors should be protected by their covers (with which they are provided), when not in use, especially if left in the open air. The larger sized tubes have a door large enough to admit the cover and so allow of its being put on the large mirror without the necessity of the speculum being removed from the body of the telescope.

To Adjust the "Finder."

Direct the telescope to any bright star (the Pole-star being by far the best, as it has very little apparent motion), and bring this star into the centre of the field of a low-power eye-piece. Now adjust the "finder" by means of the three screws bearing on it, till the star is bisected by the cross wires seen in the focus of the eye-lens of the "finder." Change the low eye-piece to a high one, and perfect the adjustment as before described. Any well-defined terrestrial object at a distance can be employed in the day-time to roughly adjust the "finder," leaving the final adjustments to be made by a star.

The use of Stops.

A stop $\frac{1}{4}$ inch less in diameter than the speculum, is often useful, in order to cut off the internal reflections of the tube.

With mirrors 6-1/2 inches in diameter and under stops are seldom required; with the 6-1/2 inch a 5 inch stop may however sometimes be used with advantage on very bright objects. With an 8-1/2 inch mirror a 7 inch stop often improves planetary definition.

With larger sizes than 8-1/2 inches several stops of different diameter may be used, when experiment will determine what size is best suited to the condition of the atmosphere, and the character and brilliancy of the object to be observed. These stops can be easily cut out of thin cardboard and afterwards blacked with Indian-ink or lampblack.

On Observing the Sun.

No larger aperture than 4-1/2 inches should be employed for solar observation, except with a specially constructed “Solar Eye-piece,” as even a 4-1/2 inch speculum often concentrates sufficient heat to crack the coloured glass of an ordinary sun-cap, if exposed for any length of time.

If a “solar eye-piece” is employed, the whole aperture may be used. In viewing the moon a slightly tinted glass is often most useful, especially to persons with weak sight, as it takes off a great deal of the glare.

TO SILVER AND POLISH THE SPECULA.

The cost of silvering is trifling, and with cleanliness and ordinary care very little difficulty will be experienced. The apparatus and chemicals required consist of the following articles:—

APPARATUS.—A Silvering Vessel:—This should be a flat-bottomed circular glass or glazed dish, 1 inch or more larger in diameter than the speculum to be silvered, and sufficiently deep to allow of a stratum of fluid of an inch or rather more between the face of the mirror and the bottom of the dish, the top of the mirror being nearly level with the edge of the dish.

A Mixing Vessel:—A 40 oz. glass measure will answer well, but care should be taken not to scratch the sides whilst stirring, or the glass is liable to fly. Should a measure not be procurable, any receptacle of sufficient size may be

used, but a glass one is best, as it will allow of the action of ammonia being better observed.

A Box of Scales and Weights.
A Glass funnel and filtering paper.
Two Glass Rods for Stirring purposes.
A Five-ounce Glass Measure.
A Test tube $\frac{3}{4}$ or 1 inch in diameter.
Some Clean Cotton Wool.
Some very fine Wash-leather.
A Support of Wood, on which to cement the speculum,
described further on.

All these articles should be perfectly clean and free from dust, and the Glass ones well rinsed with distilled water just before using.

CHEMICALS.—Nitrate Silver.

Potash pure, precipitated by Alcohol.
Sugar of milk powdered.
Nitric Acid, pure.
Liquor Ammonia.
Distilled water, pure.
Pitch.
Fine Rouge.
Turpentine.

Procure a strip of wood an inch or so less in diameter than the mirror, and of sufficient length to rest securely on the opposite sides of the silvering vessel. Pour on this piece of wood some melted pitch, and whilst it is still hot, place on it the back of the speculum moistened with a little turpentine; when cold, reverse, and lay the cemented speculum face downwards, suspended in the dish. Should the distance between the face of the mirror and the bottom of the dish be less than an inch, raise the mirror by means of thin wedges, placed between the strip of wood and the edges of the dish. But if, on the contrary, the mirror should be considerably more than an inch away from the bottom, which will occur if the silvering vessel is very deep, the speculum must be cemented to a block of wood of sufficient height, screwed to the suspending strip, instead of being cemented directly to the strip itself.

When the speculum is properly placed in the dish, namely, with the front surface about one inch from the bottom, pour in water till the fluid reaches about 1/4 inch up the side of the mirror. Measure this quantity, as it will indicate the total amount of the silvering solution required to be prepared.

To prepare the Silvering Solutions—make 3 standard solutions as follows—

- | | |
|--------|--|
| No. 1. | { Nitrate Silver in crystals 100 grains. |
| | { Distilled water 4 oz. |
| No. 2. | { Potash pure by alcohol 1 oz. |
| | { Distilled water 25 oz. |
| No. 3. | { Powdered sugar of milk 1/2 oz. |
| | { Distilled water 4 oz. |

Solution No. 3 must be made just before using. The others will keep if the distilled water employed in their preparation is pure, and the solutions, when made, are kept in glass stoppered bottles.

Suppose it is desired to silver an 8-1/2 inch mirror, proceed as follows:—the total amount of solution required having been ascertained, as before described, pour 2 ozs. of solution No. 1 into the mixing vessel previously well washed and rinsed with distilled water, and cautiously add Liquor Ammonia. A grey precipitate will be formed; continue to add the ammonia, drop by drop, till the precipitate is just dissolved and the solution becomes clear. The solution should be well stirred with a glass rod whilst adding the ammonia. Now add 4 oz. of solution No. 2, and re-dissolve the brown precipitate which is produced with Liquor Ammonia as before described. There will now be about 6-1/2 oz. of solution. Subtract the 6-1/2 oz. from the total amount previously ascertained to be required, and the remainder will be the amount of distilled water to be added. Add half this quantity of distilled water to the 6-1/2 oz. of solution, and add a drop at a time of solution No. 1, till there is a slight precipitate, which cannot be re-dissolved by a considerable amount of stirring (say for 2 or 3 minutes); then add the remaining half of distilled water and cover up from dust the vessel containing the solution, so as to allow the slight precipitate to settle. There is a point of importance to be attended to, namely, that no more ammonia is employed than is absolutely necessary. The total amount of ammonia required in the 6-1/2 oz. of solution is about 2 drs.

To Clean the Mirror.

Fill the end of a test tube with cotton wool, leaving plenty outside the tube. Having poured a small quantity of strong nitric acid on the front of the mirror, rub the acid well all over the front and sides with the cotton wool brush. Place the speculum under a water tap for a few minutes till the acid is washed away, and finally well rinse with distilled water; then place it in the silvering vessel (previously thoroughly cleansed) and pour in distilled water till it reaches 1/8 inch up the side of the speculum.

To Mix the Solutions.

The precipitate having settled, pour into a clean vessel all that is clear of the solution, leaving about 2 oz. behind, which will be turbid with the precipitate, and therefore useless. The total amount will be afterwards made correct by the addition of the same quantity of solution No. 3. Having filtered solution No. 3, warm it to about 100° F. by allowing the bottle to stand in warm water, or by heating in a small flask. When everything is quite ready, add 2 oz. of the filtered solution No. 3, whilst warm, to the clear solution described above, and thoroughly mix.

To immerse the Mirror.

Remove the mirror from the distilled water, taking care not to touch the surface of the mirror, and wipe the back and edge with some clean cotton wool. If this precaution is not taken the water is liable to drain down the sides of the mirror whilst silvering, and cause streaks at the edge of the film. Having poured away the distilled water in the silvering vessel, substitute the mixed solutions, and directly the solution becomes slightly inky, gently immerse the mirror, taking care that no air bubbles, or specks of any kind, remain between the surface of the mirror and the solution. The mirror should not be removed from the bath until all the silver has been exhausted from the solution. This may be known by the solution being clear below the silver film on the surface of the liquid. The time required will vary from 45 minutes on a hot summer's day, to 90 minutes when the thermometer shows

a low degree of temperature. In the latter case it is better to have the silvering bath in a warm room.

Immediately the mirror is removed from the bath, the silvered surface should be well washed by allowing ordinary water to flow on it from a tap, for five minutes or more, then finally rinse with distilled water and place the mirror to dry with the silvered surface resting on some blotting paper. It is as well if the mirror can be left undisturbed for a day or two, as the film will be firmer, but it may be polished, if desired, after drying for a few hours.

To Polish the Silvered Surface.

Make a couple of polishing pads by filling two pieces of very soft wash-leather about six inches square loosely with cotton wool, and tie them into balls. Gently remove any dust that may have settled on the film with some loose cotton wool, and then go over it with one of the pads in small circular strokes for about 15 minutes. This will consolidate the film and fit it for polishing. Spread a little of the finest rouge on a sheet of writing paper, and impregnate the other pad with it. Go over the film with the rouged rubber with the same circular strokes till it is perfectly polished, which will take another 15 minutes or so. Never commence with the rouged pad, as the surface may be injured. When once the film has been consolidated it will remain so, and can be repolished many times with the rouged pad should it get tarnished. The pads should be kept from dust in wide-mouthed bottles for future use. With care the film will last for a long time, especially if it is not allowed to get damp, and consequently the mirrors should never be brought uncovered from the cold air to a warmer temperature.

The “flat” may be silvered and polished in the same way as the speculum, using a smaller appropriate vessel for the silvering solution.

To Separate the mirror from the Wooden Support.

Should the mirror be attached directly to its support, insert a chisel between them, when one or two gentle blows will cause them to separate, but, should the mirror be cemented to a block, stand the mirror on edge, when a slight tap on the block will detach it. Scrape off any pitch that remains on the back

of the mirror, using finally some turpentine to wipe it clean. Great care should be taken not to finger the film.

Martin's process for Silvering.

Prepare four solutions of any quantity. Keep in stoppered bottles.

Solution 1.—Dissolve 175 grains of pure nitrate of silver in 10 oz. of distilled water.

Solution 2.—Dissolve 262 grains of pure nitrate of ammonia in 10 oz. of distilled water.

Solution 3.—Dissolve 1 oz. avoirdupois of pure caustic potash (prepared by alcohol) in 10 oz. of pure distilled water.

Solution 4.—Dissolve 1/2 oz. avoirdupois of pure sugar candy in 5 oz. of distilled water, then add 32 grains of tartaric acid and boil in a flask or other clean glazed vessel for 10 minutes, when cool add 1 oz. of alcohol and then dilute with distilled water, so as to make up the volume to 10 oz.

For silvering use equal parts of each, mix solutions 1 and 2 together, and 3 and 4; when the mirror is ready mix the whole together in the silvering vessel and quickly suspend the mirror.

In the summer time, if the solution cannot be kept in a very cool place, the mirror must be quite ready to be placed in the bath, as the solutions turn instantly when mixed together.



SELECTED TESTIMONIALS

11, WELLINGTON PARK TERRACE, BELFAST,

July 7th, 1876.

DEAR SIR,—The night before last was a clear night and I got the 6-1/2 inch out, and though the air was not good when using a 3-1/4 inch refractor, I was much pleased with the performance of the mirror.

Mr. W. came up about 11 o'clock and stayed till one, we got it on Saturn, and although rather low it was really a fine sight, Mr. W. was much pleased.

The moon being near to full we could not do much with faint points of light.

Yours sincerely, W. E. PARKINSON.

Mr. G. CALVER.

74, HAGLEY ROAD, EDGBASTON,

November 27th, 1876.

DEAR SIR,—I have now got the 5-inch speculum fairly into adjustment, and I am well pleased with it. Last night, the sky being clear for a short time, I turned it on the moon, the definition of minute craters was all that could be desired. I hope to have better opportunities.

I am, Yours truly, F. G. L.

Mr. G. CALVER.

74, HAGLEY ROAD, EDGBASTON,

November 21st, 1876.

DEAR SIR,—I have mounted the 5-inch speculum on a simple equatorial and it works well. I am sure it will turn out a fine glass and I shall not regret the time and pains I have spent over it. It is not quite in adjustment yet, when it is it will afford me a treat. I have just tried it once on the Orion Nebula and Trapezium, details of Nebula well seen, and the 5th star in Trapezium easy.

I am, Yours truly, F. G. L.

DOWLAIS,

December 27th, 1876.

DEAR SIR,—I have thoroughly tested the speculum, and am pleased to tell you it stood its trials well.

Yours truly, D. C. C.

Mr. G. CALVER.

ST. DENIES, SOUTHAMPTON,

August 14th, 1876.

DEAR SIR,—With reference to your enquiry as to the performance of the 6-1/2 in. reflector. I have pleasure in stating that I am perfectly satisfied. It readily divides the test objects and shows delta Cygni with as low a power as 160. The comes to Sirius may be considered an atmospheric test, perhaps, but I have repeatedly seen it. Your stand I find very steady and convenient; altogether I can fairly say that I consider that I have an instrument of considerable power at a comparatively small cost. I now find the attempt to observe with a refractor the reverse of pleasant.

The owner of a 3-inch refractor, after using my reflector, writes that he envies me its power, ease, definition, and comfort in observing.

I am, Dear Sir, Yours truly, A. H. S.

HILGAY RECTORY,

February 20th, 1877.

DEAR MR. CALVER,—The sky last night cleared up in places and enabled me to try the 10-inch; there was a slight haze, but I never saw discs so perfect

and neat. I felt certain it would divide anything divisible. Clouds came up and put an end to work.

I am, Dear Sir, Yours sincerely, St. V. B.

25, HAMILTON TERRACE, ST. JOHN'S WOOD,

May 26th, 1877.

MY DEAR SIR,—I have much pleasure in informing you that the very first time I saw Saturn through your 6-1/2-inch reflector, the definition of the planet was far superior to anything I had observed before, even with a good 4-1/4-inch refractor; and on another occasion I was greatly pleased with the clear and easy view of the “Comes” to ϵ Boötis. But even without including the stars, the views of Saturn and Jupiter through my 6-1/2-inch mirror do *alone* (in my opinion) well repay the cost of the telescope.

Yours truly, WM. L. LANCASTER.

Mr. G. CALVER.

37, EATON RISE, EALING,

February 13th, 1877.

DEAR SIR,—I like the 18-inch speculum, and I think it as good as it can be, and if the larger one is as good I shall be delighted with it. It gives beautiful star images: I see Sirius as a brilliant dot, a glorious object without ray or flares of any kind. I never saw it so well before.

Yours truly, A. A. C.

Mr. G. CALVER.

37, EATON RISE, EALING,

February 3rd, 1877.

DEAR SIR,—I have tried the 18-inch speculum on some tests, and especially the satellites of Uranus, and it appears fine, very fine. What I have done in photography promises well.

Yours truly, A. A. C.

Mr. G. CALVER.

SOUTHEND ON SEA,

March 17th, 1877.

MY DEAR SIR,—Circumstances have prevented me from making many observations of late, but I am more confirmed in my opinion that my telescope of your make is a very fine instrument. (A 6-1/2-inch.) I heartily wish you success, which I feel sure you will obtain, as you are so painstaking and turn out a thoroughly reliable article. I shall be curious to compare my brother's telescope with mine.

Sincerely yours, J. L. L.

Mr. G. CALVER.

GORLESTONE,

March 30th, 1875.

DEAR SIR,—You have asked me for my opinion of the 10-inch speculum. I have tested it, and can speak in the highest terms of its performance. Definition in good air is as near perfection as it is possible to imagine. Such tests as ζ Cancri, γ^2 Andromedæ are well divided. The 6th star in the trapezium stands out well clear of its brighter neighbour. On the moon any power may be applied, only lessening the light, but retaining sharpness of outline. I have tried a good many telescopes, but never yet saw these 10-inch

mirrors surpassed. The image of a star disc expanded on each side of the focus is of as nearly the same appearance as possible.

Yours truly, W. P. MATTHEWS.

416, BRIXTON ROAD,

April 20th, 1875.

DEAR SIR,—In compliance with your request, I send particulars of my observation on ϕ Draconis. It was about 12.45 this morning when I directed my 8-1/2-inch to this object, the air was very unsteady, in fact not nearly so good as some nights lately, and I well divided it with a power of 450. I then used a 6-1/2-inch stop, which I found very much increased the steadiness of definition.

Yours very respectfully, P. H.

10th July, 1874.

DEAR SIR,—I tried the 6-1/4-inch mirror I had from you for the first time last night. The night was not a good one for definition, but I was very much pleased indeed with its performance.

Yours truly, C. H. W.

HONITON RECTORY,

Dec. 23rd, 1874.

DEAR SIR,—I have been wishing to tell you how pleased I am at the performance of your mirror. Many thanks for your kind offer to change the mirror if needful, but I think it could hardly be better than it is, its performance on different double stars is most excellent. I have examined

nearly 100 of these since September with the 6-1/2-inch, most of them very difficult, but the mirror came out well under all tests. I send you a list of some of the objects I have examined.

Yours truly, H. SADLER.

π *Aquilæ*.—Divided, power 80. 14^m. Webb might have been rated 15^m. [6^m, 7^m. 1"·5.]

δ *Aquilæ*.—[A 3-1/2, B 16, C 14, A-B 96", A-C 194".] Comites easy, 3 other excessively minute ones, not shown by the 11-in. refractor with which P. Smyth measured B.C. Found 6-1/2-in. Sept. 12th, est. $\pm$ 17 mag. 100" to 110".

ζ *Persei*.—My friend "Linea" sees two minute stars with 4·28-in. Wray, not in Webb, and Wray himself sees an excessively difficult companion. I see these 3 stars easily with 6-1/2-in. (Wray says his companion is "very difficult with 7-in. refractor, really only a glimpse star, even with the best atmosphere"), and have added two more stars to the group.

β *Delphini*.—Close double discovered by Burnham, est. 0"·7. In contact 6-1/2-in., power 430.

η *Coronæ Borealis*.—Very low, long past meridian, in contact 164 power.

β *Equalei*.—[a 5-1/2, b 13, c 14, B 16: a-b 35", a-c 50", b-B 3".] 16^m. not seen, Webb 9-1/3-in. Just divided 164, easy 430 power. Two other companions (not seen by Smyth or Webb?)

μ *Andromedæ*.—The 16^m. comes, a very difficult test, easy; other comites (not seen by Smyth or Webb?) seen.

Companion to Vega pretty easy, Nov. 12th, 37m. after sunset.

P. 178 xx. *Delphini*.—[a 7-1/2, b 8, c 16, B 9: a-b 14"·3, b-B 0"·7, a-c 20".] Smyth 16 by evanescent glimpses; easy 164 power, 6-1/2-in. 8^m. well elongated in direction of 230°, power 430.

REGENT ROAD, GREAT YARMOUTH,

October 24th, 1874.

DEAR SIR,—Doubtless you have anticipated hearing from me, relative to the 10-inch telescope, but the weather has been so indifferent for delicate astronomical work that it has not yet had a fair trial.

In middling good air, when I could use a reduced aperture with satisfactory results, the star images reminded me of the 6-1/2-inch, which I considered so superb a mirror that nothing could surpass its fine performance either on stars or planets.

With the recollections of the truly splendid views I had of Jupiter last season, I count much to see him with the 10-in.

Yours truly, H. BLYTH.

REGENT ROAD, GREAT YARMOUTH.

February 2nd, 1875.

DEAR SIR,—In reply to yours, I have not had a really good night to test severely the defining powers of the 10-inch mirror, but judging from the few difficult objects I have seen, the mirror promises well, and I have no doubt of its excellence.

I had a fine view of the grand nebula in Orionis, and noticed that the 5th and 6th stars were quite plainly seen, when the aperture was reduced to 5-1/2-inches. I consider your stand a great improvement, being very convenient and steady.

Yours truly, T. AYERS.



CATALOGUE
OF
Silvered Glass Reflecting
Telescopes, &c.,
AND THEIR ACCESSORIES,

JUNE, 1877.

G. Calver, while introducing the list of prices below, has the greatest confidence in calling the attention of the practical astronomer and the amateur to the very moderate prices charged, considering the principle and style of the mounting, and the firm and well-fitted arrangements in all the parts.

The principle of the mounting is that which is the most convenient to use, and with the greatest degree of steadiness with the easiest movements. Fig. 1 is the *best mounting ever applied to the Reflecting Telescope*.

He is also able to state, that with his processes and special facilities for working large specula, and substantially mounting them, he is prepared to construct automatic equatorials of large sizes, and with any special arrangements that may be required.

SILVERED GLASS EQUATORIAL TELESCOPES, very substantially and well-fitted, as (Fig. 2). All these equatorials have revolving body.

£ s. d.

5-1/4 inch Speculum from 4 to 6 feet focus, with 8 inch hour
circles reading to 5 seconds of time, and declination circle
reading to 1 minute, two powers

30 0 0

6-1/2 inch Speculum of from 5 to 6-1/2 feet focus, 10 inch rotating hour circle, reading to 5 seconds of time, and declination circle to 1 minute of arc, with 3 powers—100 to 500	45 0 0
8-1/2 inch Speculum as above with 12 inch circles, 4 powers	70 0 0
9-1/4 inch Speculum as above with 12 inch circles, 5 powers	78 0 0
10 inch Speculum as above with 12 inch circles, 5 powers	90 0 0
12-1/2 inch Speculum as above with 12 inch circles, 5 powers	120 0 0

Any of the above sizes mounted on Stand (Fig. 1), from 5 per cent. extra.

Fig. 1 is a very excellent stand, and admirably suited for large instruments with clock power, and is convenient to remove and adapt to different latitudes, &c., see [p. 20](#).

£ s. d.

14, 15, to 16 inch Speculum mounted as (Fig. 1), with rotating hour circle reading to 5 seconds and declination to 1 minute (the diameter of the circles never less than that of the Speculum and often larger)	
8 powers from 50 to 700, with first-rate and very powerful driving clock, first-class instrument, and complete	from £220 to 330 0 0
17, 18, to 20 inch Speculum, with 10 powers, from 50 to 800, including Kellners, Huyghenians, and Achromatics,	from £350 to 500 0 0
22 to 24 inch Speculum	from £550 to 750 0 0
30 inch Speculum, with 12 powers and position micrometer, and transit eye-pieces	1000 0 0

Prices will be forwarded for special arrangements in any of these large sizes.

The CASSEGRAIN TELESCOPE is a very convenient form, and perfect in performance, and in this form a very large instrument requires comparatively a much smaller observatory, and is more conveniently worked. They can be made so that the image is thrown out at the side and viewed diagonally, as with the Newtonian, see [p. 35](#). Cassegrains from 5 per cent. extra.

		£ s. d.
Driving Clocks for	6 to 10 inch	35 0 0
	12 18	40 0 0
	20 30	50 0 0

The EDUCATIONAL REFLECTOR is a plain and very steady and satisfactory instrument, mounted on (Fig. 2) stand, without circles, has revolving body, and made so as to be portable.

	£ s. d.
6-1/2 inch Speculum (clear aperture), and of 4 to 6-1/2 feet focus, has screw-motion in right ascension, 2 powers	25 0 0
8-1/2 inch speculum as above, with 3 powers	40 0 0

Any of these equatorials can be, according to wish, fitted without the circles, &c., and dispensing with that which is not needed, so as to reduce the price to suit circumstances and convenience.

The POPULAR REFLECTOR, with Angle-Block Stand, with endless screw-motion to follow the stars with equatorial motion.

	£ s. d.
5-1/4 inch Speculum, of from 4 to 6 feet focus, with 2 powers	15 15 0

(See [p. 20, Fig. 4](#)).

Reflecting Telescope on Alt-azimuth Stand, as described at [page 13](#), fitted with Silvered Glass Speculum, and provided with Two Eye-pieces.

		£ s. d.
5-1/4 inch Speculum, 4 to 6	feet focus	19 10 0
6-1/2	4 6-1/2	24 0 0
8-1/2	6-1/2	33 10 0
10	7	45 0 0
12-1/2	8	70 10 0

Reflecting Telescope on Cradle Stand, and provided with Two Eye-pieces.

		£ s. d.
5-1/4 inch Speculum, 4 to 6 feet focus		21 0 0
6-1/2	4 6-1/2 or 5 feet focus	27 0 0
8-1/2	6-1/2	36 0 0
10	7	49 0 0
12-1/2	8	76 0 0

SILVERED-GLASS SPECULA (*unmounted*).

THE FINEST QUALITY GUARANTEED.

		£ s. d.
5-1/4 inch diameter 4 to 6 feet focus		5 0 0
6-1/2	4 6-1/2	6 5 0
8-1/2	6-1/2	13 0 0
10	7	25 0 0
12-1/2	8	38 10 0
14	8	55 0 0
16	10	75 0 0
18	12	100 0 0
20	12	130 0 0
22	15	165 0 0
24	18	200 0 0
26		250 0 0
30		300 0 0
36		400 0 0

SILVERED-GLASS DIAGONAL MIRRORS (*unmounted*).

FINEST QUALITY GUARANTEED.

	£ s. d.
1 inch in the minor axis, or narrowest diameter of the ellipse	1 0 0

1-1/2	ditto	ditto	ditto	1 10 0
2	ditto	ditto	ditto	2 0 0
2-1/2	ditto	ditto	ditto	2 10 0
3	ditto	ditto	ditto	3 3 0
3-1/2	ditto	ditto	ditto	4 4 0
4	ditto	ditto	ditto	5 5 0

SILVERING AND POLISHING SPECULA.

	£	s.	d.
5 inch	0	6	0
<u>6-1/2</u>	0	8	0
<u>8-1/2</u>	0	10	6
10	0	15	0
12-1/2	1	1	0
Diagonal Planes 3s. to	0	5	0

ASTRONOMICAL EYE-PIECES.

OF BEST QUALITY.

Huyghenian Construction of the following magnifying powers on a 6-1/2 feet focus object-glass:—

	£	s.	d.
35, 60, 90, 130, 180, 200	each	0	15 0
320, 450		1	1 0
610		1	7 6

All fitted with Sunshades.

Improved Achromatic, Ramsden's Construction.

	£	s.	d.
90, 130	each	1	0 0

189, 250	1 10 0
320, 450	1 12 6
610, 750	1 15 0
850	2 2 0

Kellner's Construction.

	£ s. d.
30, 58, and 80	each 1 5 0

SOLAR EYE-PIECES.

	£ s. d.
Solar Diagonal, with true plane	1 12 6
Dawes' Solar Eye-piece	8 0 0
Barlow's Lens	1 1 0
Day Erecting Eye-piece	1 5 0
Coloured Eye-piece Cap for observing the Moon	0 4 0
Transit Eye-pieces, for use with Equatorials	£1 1s. to 2 2 0
Adjusting Piece	0 2 6

MICROMETERS.

	£ s. d.
Parallel Wire Micrometer	5 5 0
Position Micrometers	£8 8s. to 15 15 0
Double Image Micrometer	8 8 0
Illuminating Apparatus	from 8 0 0

Good Instruments will be taken in exchange and liberally allowed for.

GEORGE CALVER,
HILL HOUSE, WIDFORD,

CHELMSFORD, ESSEX.



APPENDIX
TO
**Hints on Silvered Glass
Reflecting Telescopes.**



he object of this pamphlet is to give some new and additional advice not contained in my “Hints on Silvered Glass Reflecting Telescopes,” on silvering and adjusting them; and, as the results of continued and recent experience, it is hoped they may be useful, and may prove an acceptable appendix to my little book of Hints.

The first thing to do after fastening the mirror to the wooden support, is to suspend it in the dish in which it is to be silvered, and so to adjust it that there shall be one inch between the bottom of the mirror and the bottom of the dish; then pour in water to come a quarter of an inch up the sides of the glass—the quantity thus found is to be measured, and will be the exact amount of the bath when all the solutions are mixed; this will prevent any hitch at an important moment, and the glass can be immersed without delay or disturbance.

Next proceed to wash the surface with nitric acid, taking care that the acid does not run down the sides, as it is not so easily removed from the fine ground sides as it is from the polished surface. After gently but thoroughly rubbing the surface, add a little water, and again go over; then wash all off, and take a large piece of cotton wool and well sponge the surface and sides with plenty of water, and suspend in a dish or plate with water in it.

TO PREPARE THE SOLUTIONS.

The plan I adopt is this,—I dissolve a large quantity of the chemicals required in one-fourth or one-eighth the quantity of water employed by Martin, so that in a Winchester quart stoppered bottle I can keep a large

supply ready. I make the silver and ammonia solutions eight times the original strength, the potash and sugar solutions four times only, as these latter solutions require more water to properly prepare them.

Enough solutions to silver a 6-1/2 in. mirror, eight times are to be made.

Dissolve 1400 grains of nitrate of silver in 10 ounces of water (it may be clean fresh rain water filtered, if distilled is difficult to obtain, and it will act very well), but do not put the silver into the exact 10 ounces, but make up to exact 10 ounces after the silver is dissolved. It is thus condensed eight times. Do the same with the nitrate of ammonia, by dissolving 2096 grains.

Next prepare the potash and sugar solutions, condensed to four times the initial strength.

Dissolve 8 ounces of potash, and make up to exactly 20 ounces of water; if this is dissolved in a glass measure it will evolve sufficient heat to break it; it is safely done in a clean white jug. Lastly, dissolve 4 ounces of white sugar candy with 416 grains of tartaric acid, and boil ten minutes in a clean glazed vessel; when cold, add 8 ounces of alcohol, and make up to 20 ounces with water.

Do not use the solutions till all are of the same temperature, nor on the same day as they are made; and do not silver until the glass, and *everything* used are of the same temperature. To insure this, get the mirror ready and suspend in water, with the solutions all collected in the room in which the silvering is to be done, and let them remain until next day.

Now, remembering how many ounces were needed to leave one inch of solution under the surface and a quarter of an inch up the sides of the 6-1/2 in. mirror, using a vessel about 2 inches larger in diameter than the mirror, pour into a glass measure 10 drams of the silver solution, add next 10 drams of the ammonia, then 20 drams of potash; if the potash turns the mixture thick, pour it backwards and forwards into another vessel, or stir it with a strip of glass, or a glass rod, for half a minute; if the mixture does not turn clear (which it will not do if it has turned very thick on adding the potash), add cautiously, drop by drop, some of the ammonia solution, agitating it till it just clears; do not filter unless there are a great many floating particles through using unfiltered water, the floating particles cause minute black spots, which are, however, of little consequence, as their action is only the loss of so much light. Measure 20 drams of the sugar—and the mirror being

ready to dip (the amount with the sugar added must be the required quantity previously ascertained), add the sugar, stir well, and immerse when the mixture begins to turn dark ink colour.

When silvered, well wash the mirror with water of the *same* temperature as itself (it may be safely sponged with a lump of cotton wool), and stand it on its edge on blotting-paper to dry. Drops of water standing long on it when drying will cause stains, which will not readily polish off, and indeed these should not be polished, as it is not right to polish one part more than another; so it is best to leave them, or if this is objected to, re-silver the mirror, when standing it in the sun and wind will soon dry the surface, but be careful not to let the dust settle on it.

Do not warm anything; if the weather is cold, everything may be left many hours in a warm room, but the *equal* temperature is of the utmost importance—the actual temperature is not of much consequence. More failures are due to unequal temperature than any other cause in the hands of the amateur. One careful and clever amateur informed me that he had failed six times, but on the seventh, by attending to the advice given above, the silvering was a perfect success, and all seemed then as easy and certain as it had before seemed uncertain and difficult.

THE FINAL ADJUSTMENT OF A NEWTONIAN REFLECTOR ON A STAR.

The adjustments being as near correctness as can be seen by the usual method of procedure—of which the fullest details are given in my little book, the telescope is ready to try on a star. If all does not appear satisfactory on first observing the star, do not disturb the adjustments for a while, but wait until the air is steadier, when perhaps it will be quite satisfactory. If not, turn it on a bright star; if the adjustments are at fault, the black shadow of the flat, when the star is out of focus, will not be central in the slightly expanded image, but will be on one side of the expanded disk. Use a power of 200 to 300.

If the black spot is near one of the sides corresponding with major axis of flat, the side screw will put it right. We will assume that the black spot is nearer the edge of flat nearest the mirror, and that the eyepiece is beyond the focus proper for the star; then *unscrew* the side screw. If the spot is nearer

the other edge of flat, *screw it up* little by little, taking the hand out of the telescope each time, and see what the effect is. The movement and alteration can be watched while the hand is in the tube. If the spot is seen either the top or bottom of the flat that is in the direction of the diameter of tube, the middle screw must be slightly loosened and the flat revolved. Let the screw just bite so as to hold, and tighten it when correct. The definition of a star should now be perfect, and by daylight observe the adjustments; it may appear that the reflected circles are not concentric, but if the definition on a star is good in still air, consider the appearances correct adjustments, except that the reflection of the black spot on mirror may be made concentric. It sometimes happens that a persistent and rigid centreing of the reflections on *face of flat* are not the positions for best definition, perhaps from a slight optical eccentricity somewhere in the instrument.

The adjustments of a reflector are soon familiar and easy, and there is this to be said, the adjustments can be manipulated upon to any amount without the slightest injury to the instrument; there is no danger of any sort, and the instrument can be perfected in its adjustments by daylight, and this makes it both pleasant and convenient to leisurely work at, for experience sake. The instrument need not be out of doors either, unless an artificial star from a black pin's head—which is better in *strong* sunlight than a thermometer bulb—is to be used instead of a star at night, if so let it be as high as possible.

Be careful not to screw up the small screws too tight, as the fine threads are liable to be spoiled and the screw made useless.

THE DURABILITY OF SILVER FILMS.

It sometimes happens that the films do not last so long as they are expected to do, that is, they sometimes lose their splendid lustre sooner than they should do. When it is remembered that though the process of silvering is an easy operation to perform, yet it is a delicate chemical one, in which good results are best obtained when several good conditions all meet together, this is not to be wondered at, but it is certain the instructions given above will reduce the chances of failure to a minimum, and imperfect results will be rare. The silver usually lasts a long time. I know silver films that have been in use for ten years.

Do not suppose that a long and badly worn surface will fail to show the detail and colours of the planets, the fullest detail also of the moon or the stars. It is surprising how long a silvered surface will continue to do its full work while its appearance is much deteriorated; picking up faint points of light that are near the very limit of its aperture and power, seem alone to require the perfect and fresh film. I know silver films that have been in use for seven years and have not been ever re-polished.

That the Silvered Glass Reflecting Telescope is giving satisfaction and is capable of performing the best work of a telescope—being durable, convenient, and perfectly efficient—may be gathered from the selection of a very large number of satisfactory and most gratifying letters received by the writer; and it has been most encouraging to find in all cases the observer is most eager to express his entire satisfaction, and especially so where so many have been prejudiced, in the absence of experience, with feelings of doubt as to their *complete* efficiency, and many, again, having used fine refractors.

It will be seen, too, that many have commenced with a small sized reflector, and gone on increasing their optical power until they have obtained a large and powerful instrument, imposing indeed as compared with what was considered a powerful instrument less than half a century ago, and at a fraction of its cost.

The gratifying success which I have achieved, attested by the universal satisfaction given by the instruments supplied by me, is the result of unwearying labour and untiring patience; every speculum, large or small, being figured with my own hands.

I have every facility for setting up instruments, and for testing, working, and regulating them in every part on celestial objects. The optical parts are, therefore, not merely tested separately and subsequently mechanically adjusted, but each and every instrument is put together and optically and mechanically tested, as a whole, before it is permitted to leave the workshop.

Every speculum is most carefully figured, and confidently guaranteed perfect *to the extreme edge*.

The employment of “stops” of any kind is quite unnecessary, except on bad nights, when the aperture must of course be suited to the degree of steadiness of the air. For, whether the instrument be a reflector or a refractor, the

aperture and power used must of necessity be limited by the atmospheric conditions under which observations are made.

It should, however, be remembered that the reflector having not only a much larger aperture than a refractor of same focal length, and, being open to the influence of the external air, is not only affected by a relatively larger column of air, but is affected in a different way. It frequently happens that reflectors of 6 in. to 18 in. aperture, are of same focal length as 5 or 6 inch refractors.

These considerations will explain the occasional use of stops, and the expediency of having them ready to use when they can be of temporary advantage. There are nights in our climate when a 6 inch aperture may be the largest that can be used with satisfaction or advantage; indeed it used to be accepted as an incontrovertible fact that an aperture of 6 inches was the largest that could be used on *average* nights in our country. But there are also not a few nights when large apertures, bring great gain to their fortunate possessors.

It is obvious that, *mutatis mutandis*, these conditions must affect refractors as well as reflectors, with perhaps this difference, that in the case of a reflector the rays pass down the tube and infringe on the speculum as a column of parallel rays; consequently all the external rays of the column travel near the surface of the tube for its entire length. This being so, it is obvious that until the internal and external air and the metal tube have time to equalize in temperature, it will be advantageous, in all work requiring high powers, such as the examination of difficult double stars, to “stop off” for a time the *extreme edge* by a small diaphragm.

This will occur especially on frosty nights after a warm fine day. When the telescope is in the open air, ice will not infrequently form on the top of the tube, and the effect will be at once obvious in the irregularity of a star image at the edge corresponding to the top of the tube. The use of a small diaphragm will at once obviate this defect and give perfect images. On such nights a wooden tube is preferable to a metal one. It has been frequently my experience, when finally testing an instrument, that when I have been dissatisfied with its performance, and in order to trace the cause of the apparent defect, have removed the mirrors to my wooden testing tube, I have found every defect removed, and imperfect replaced by perfect images. The explanation is simple. Not only is the wooden tube much larger in diameter

than the specula, but wood is of itself less sensible than metal to differences of temperature.

To make the tubes some 2 or 3 inches larger in diameter than the speculum would cure this evil, but it would involve various inconveniences and expenses in structure, which would more than outweigh the advantage of overcoming a hindrance which is after all only temporary and occasional.

The experiment of perforating or ventilating tubes has now been thoroughly and exhaustively tried, with the result that there is a fairly general consensus of experienced opinion against their use. The advantages are merely theoretical; the disadvantages are grave. Among them are the admission of cross reflections in every direction, the admission of dust, imperfect protection of the mirrors, and a decided loss of strength and rigidity to the tube.

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OCTOBER, 1880.

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The principle of the mounting is that which is the most convenient to use, and with the greatest degree of steadiness with the easiest movements. Fig. 1 is the *best mounting ever applied to the Reflecting Telescope*.

He is also able to state that, with his processes and special facilities for working large specula, and substantially mounting them, he is prepared to construct automatic equatorials of large sizes, and with any special arrangements that may be required.

Fig. 1 is a very excellent mounting, and admirably suited for large instruments, especially when clock-power is applied; and as now manufactured by G. CALVER, is the most complete and reliable, its details of construction being such as to give the greatest freedom of motion and steadiness. For large sizes the top of polar axis works on friction rollers; and, as a superior German stand, its arrangements and means of astronomical as well as optical adjustments, are such as to make it an instrument of precision.

To ensure these qualities many well considered arrangements—entailing careful and expensive workmanship—must be provided; details that do not easily admit of description in an ordinary catalogue, or can be shown in an engraving, but are duly appreciated by the observer, who will find their value by practice.

The cradle bar is compound with fine screw movement for perfecting collimation, but to make this adjustment perfect, the telescope tube must be centred in a powerful lathe, and the cradle and solid metal rings in which the tube revolves ([see illustration](#)) has to be turned and fitted with true flanges or working bearings; the mechanical and optical centres will then coincide.

The hour circle, to be truly divided, must be truly made, and is a solid wheel of good substance, working on a secondary axis, and rotated by mechanical means, truly concentric, with a very strong polar axis.

The declination circle has fine tangent screw movement, with double action for setting the readings by milled-headed screw, and also long driving rod from the eye-piece to set and adjust the object in centre of field. There is slow hand motion, by a separate wheel in right ascension.

The clock-power is connected and disconnected instantly, by touching with the finger a small lever, conveniently placed at a small door in the clock case. The clock will go at exactly the same rate, whether it is driving the telescope or not, and the latter begins to move at the proper speed the instant the lever is moved, and the connection of the telescope and clock made, which is instantly effected by another lever.

£ s. d.

14, 15, to 16 inch Speculum mounted as (Fig. 1), with rotating hour circle reading to 5 seconds and declination to 1 minute (the diameter of the circles never less than that of the Speculum, and often larger)

8 powers, from 50 to 700, with first-rate and very powerful driving clock, first-class instrument, and complete

from £250 to 350 0 0

17, 18, to 20 inch Speculum, with 10 powers, from 50 to 800, including Kellners, Huyghenians, and Achromatics,

from £360 to 500 0 0

22 to 24 inch Speculum	from £600 to	700 0 0
30 inch Speculum, with 12 powers and position micrometer, and transit eye-pieces		1000 0 0

Prices will be forwarded for special arrangements in any of these large sizes.

		£ s. d.
Driving Clocks for	6 to 8 inch	35 0 0
	12 18	40 0 0
	20 30	50 0 0

These are fitted with governor, regulator, and self-adjusting break. They are made of gun metal and steel, and every wheel is cut. They drive with most excellent regularity, and are, when desired, made to beat seconds on a bell. The whole is enclosed in a mahogany frame, with glass panels.

Fig. 2 admits of equal completeness, &c., but the stand is not so convenient for clock-power.

SILVERED GLASS EQUATORIAL TELESCOPES, very substantially and well fitted as (Fig. 2). All these equatorials have revolving body.

	£ s. d.
5-1/4 inch Speculum, from 4 to 6 feet focus, with 7 inch hour circles, reading to 5 seconds of time, and declination circle reading to 1 minute, 2 powers	30 0 0
6-1/2 inch Speculum, of from 5 to 6-1/2 feet focus, 10 inch rotating hour circle, reading to 5 seconds of time, and declination circle to 1 minute of arc, with 3 powers—100 to 500	45 0 0
8-1/2 inch Speculum, as above, with 10 inch circles, 4 powers	70 0 0
10 inch Speculum, as above, with 10 inch circles, 5 powers	90 0 0
12-1/2 inch Speculum, as above, with 10 inch circles, 5 powers	120 0 0

The EDUCATIONAL REFLECTOR is a plain and very steady and satisfactory instrument, mounted on (Fig. 2) stand, without circles, has revolving body,

and made so as to be portable.

£ s. d.

8-1/2 inch Speculum as above, with 3 powers 40 0 0

The POPULAR REFLECTOR (Fig 3), with Angle-Block stand, with endless screw-motion to follow the stars with equatorial motion.

£ s. d.

5-1/4 inch Speculum, of from 4 to 6 feet focus, with 2 powers 15 15 0

These are also made with rotating body.

Reflecting Telescope on Alt-azimuth Stand, fitted with silvered glass Speculum, and provided with two eye-pieces.

£ s. d.

5-1/4 inch Speculum, 4 to 6	feet focus	19 10 0
6-1/2	5 6-1/2	25 0 0
8-1/2	6-1/2	33 10 0
10	7	50 0 0
12-1/2	8	70 10 0

Silvered-Glass Specula (*unmounted*).

THE FINEST QUALITY GUARANTEED.

£ s. d.

5-1/4 inch diameter, 4 to 6	feet focus	5 0 0
6-1/2	4 6-1/2	6 5 0
8-1/2	6-1/2	13 0 0
10	7	25 0 0
12-1/2	8	38 10 0
14	8	55 0 0

16	10	75 00
18	12	100 00
20	12	130 00
22	15	165 00
24	18	200 00
26		250 00
30		300 00
36		400 00

Silvered-Glass Diagonal Mirrors (*unmounted*).

FINEST QUALITY GUARANTEED.

				£ s. d.
1 inch in the minor axis, or narrowest diameter of the ellipse				1 0 0
1-1/2	ditto	ditto	ditto	1 10 0
2	ditto	ditto	ditto	2 0 0
2-1/2	ditto	ditto	ditto	2 10 0
3	ditto	ditto	ditto	3 3 0
3-1/2	ditto	ditto	ditto	4 4 0
4	ditto	ditto	ditto	5 5 0

Silvering and Polishing Specula.

		£ s. d.
5	inch	0 6 0
6-1/2		0 8 0
8-1/2		0 10 6
10		0 15 0
12-1/2		1 1 0
Diagonal Planes 3s. to		0 5 0

ASTRONOMICAL EYE-PIECES.

OF BEST QUALITY.

Huyghenian Construction of the following magnifying powers on a 6-1/2 feet focus object-glass:—

	£ s. d.
35, 60, 90, 130, 180, 200	each 0 15 0
320, 450	1 1 0
610	1 7 6

All fitted with Sunshades.

Improved Achromatic, Ramsden's Construction.

	£ s. d.
90, 130	each 1 0 0
189, 250	1 10 0
320, 450	1 12 6
610, 750	2 2 0
850	2 2 0

Kellner's Construction.

	£ s. d.
30, 58, and 80	each 1 5 0

Solar Eye-Pieces.

	£ s. d.
Solar Diagonal, with true plane	1 12 6
Barlow's Lens	1 1 0
Day Erecting Eye-piece	1 5 0
Coloured Eye-piece cap for observing the Moon	0 4 0
Transit Eye-pieces, for use with Equatorials	£1 1s. to 2 2 0
Adjusting Piece	0 2 6

Good Instruments will be taken in exchange, and liberally allowed for.

GEORGE CALVER,

HILL HOUSE, WIDFORD,
CHELMSFORD, ESSEX.

LETTERS AND EXTRACTS FROM THE FOLLOWING GENTLEMEN.

From W. E. PARKINSON, Esq.

11, WELLINGTON PARK TERRACE, BELFAST,

July 7th, 1876.

DEAR SIR,—The night before last was a clear night, and I got the 6-1/2 inch out, and though the air was not good when using a 3-1/4 inch refractor, I was much pleased with the performance of the mirror.

Mr. W. came up about eleven o'clock and stayed till one; we got it on Saturn, and although rather low it was really a fine sight—Mr. W. was much pleased.

The moon being near to full we could not do much with faint points of light.

Yours sincerely,

(Signed.)

Mr. G. CALVER.

From F. G. LEMMAN, Esq.

74, HAGLEY ROAD, EDGBASTON,

November 27th, 1876.

DEAR SIR,—I have now got the 5 inch speculum fairly into adjustment, and I am well pleased with it. Last night, the sky being clear for a short time, I turned it on the moon; the definition of minute craters was all that could be desired. I hope to have better opportunities.

I am,

Yours truly,

(Signed.)

Mr. G. CALVER.

(From the same Gentleman.)

74, HAGLEY ROAD, EDGBASTON,

November 21st, 1876.

DEAR SIR,—I have mounted the 5 inch speculum on a simple equatorial, and it works well. I am sure it will turn out a fine glass, and I shall not regret the time and pains I have spent over it. It is not quite in adjustment yet, when it is it will afford me a treat. I have just tried it once on the Orion Nebula and Trapezium, details of Nebula well seen, and the 5th star in Trapezium easy.

I am,
Yours truly,
(Signed.)

From D. C. CARMICHAEL, Esq.

DOWLAIS,

December 27th, 1876.

DEAR SIR,—I have thoroughly tested the speculum, and am pleased to tell you it stood its trials well.

Yours truly,
(Signed.)

Mr. G. CALVER.

ST. DENIES, SOUTHAMPTON,

August 14th, 1876.

DEAR SIR,—With reference to your enquiry as to the performance of the 6-1/2 inch reflector, I have pleasure in stating that I am perfectly satisfied. It readily divides the test objects, and shows delta Cygni with as low a power as 160. The comes to Sirius may be considered an atmospheric test, perhaps, but I have repeatedly seen it. Your stand I find very steady and convenient; altogether I can fairly say that I consider that I have an instrument of considerable power at a comparatively small cost. I now find the attempt to observe with a refractor the reverse of pleasant.

The owner of a 3 inch refractor, after using my reflector, writes that he envies me its power, ease, definition, and comfort in observing.

I am, dear Sir,
Yours truly,
A. H. S.

From W. L. LANCASTER, Esq.

25, HAMILTON TERRACE, ST. JOHN'S WOOD,

May 26th, 1877.

MY DEAR SIR,—I have much pleasure in informing you that the very first time I saw Saturn through your 6-1/2 inch reflector, the definition of the planet was far superior to anything I had observed before, even with a good 4-1/4 inch refractor; and on another occasion I was greatly pleased with the clear and easy view of the “Comes” to ϵ Boötis. But even without including the stars, the views of Saturn and Jupiter through my 6-1/2 inch mirror do *alone* (in my opinion) well repay the cost of the telescope.

Yours truly,
(Signed.)

Mr. G. CALVER.

Footnote:

[1] This alludes to the 37 inch, then under consideration; and I may now add that Mr. Common is perfectly satisfied with it, and “consider it a *very fine* mirror.”

(From the same Gentleman.)

37, EATON RISE, EALING,

February 3rd, 1877.

DEAR SIR,—I have tried the 18 inch speculum on some tests, and especially the satellites of Uranus, and it appears fine, very fine. What I have done in photography promises well.

Yours truly,

(Signed.)

Mr. G. CALVER.

From J. L. LANCASTER, Esq.

SOUTHEND-ON-SEA,

March 17th, 1877.

MY DEAR SIR,—Circumstances have prevented me from making many observations of late, but I am more confirmed in my opinion that my telescope of your make is a very fine instrument (a 6-1/2 inch). I heartily wish you success, which I feel sure you will obtain, as you are so painstaking and turn out a thoroughly reliable article. I shall be curious to compare my brother's telescope with mine.

Sincerely yours,
(Signed.)

Mr. G. CALVER.

From The REV. W. P. MATTHEWS.

GORLESTONE,

March 30th, 1875.

DEAR SIR,—You have asked me for my opinion of the 10 inch speculum. I have tested it, and can speak in the highest terms of its performance. Definition in good air is as near perfection as it is possible to imagine. Such tests as ζ Cancri, γ^2 Andromedæ are well divided. The 6th star in the Trapezium stands out well clear of its brighter neighbour. On the moon any power may be applied, only lessening the light, but retaining sharpness of outline. I have tried a good many telescopes, but never yet saw these 10 inch mirrors surpassed. The image of a star disc expanded on each side of the focus is of as nearly the same appearance as possible.

Yours truly,
(Signed.)

416, BRIXTON ROAD,

April 20th, 1875.

DEAR SIR,—In compliance with your request, I send particulars of my observation on ϕ Draconis. It was about 12.45 this morning when I directed my 8-1/2 inch to this object, the air was very unsteady, in fact not nearly so good as some nights lately, and I well divided it with a power of 450. I then used a 6-1/2 inch stop, which I found very much increased the steadiness of definition.

Yours very respectfully,

P. H.

July 10th, 1874.

DEAR SIR,—I tried the 6-1/4 inch mirror I had from you for the first time last night. The night was not a good one for definition, but I was very much pleased indeed with its performance.

Yours truly,

C. H. W.

From H. SADLER, Esq., F.R.A.S.

HONITON RECTORY,

December 23rd, 1874.

DEAR SIR,—I have been wishing to tell you how pleased I am at the performance of your mirror. Many thanks for your kind offer to change the mirror if needful, but I think it could hardly be better than it is, its performance on different double stars is most excellent. I have examined nearly 100 of these since September with the 6-1/2 inch, most of them very difficult, but the mirror came out well under all tests. I send you a list of some of the objects I have examined.

Yours truly,

H. SADLER.

π *Aquilæ*.—Divided, power 80. 14^{-m}. Webb might have been rated 15^{-m}. [6^{-m}, 7^{-m}. 1"·5.]

δ *Aquilæ*.—[A 3-1/2, B 16, C 14, A-B 96", A-C 194".] Comites easy, 3 other excessively minute ones, not shown by the 11 inch refractor with which P. Smyth measured B.C. Found 6-1/2 in. Sept. 12th, est. $\pm$ 17 mag. 100" to 110".

ζ *Persei*.—My friend "Linea" sees two minute stars with 4.28 in. Wray, not in Webb, and Wray himself sees an excessively difficult companion. I see these 3 stars easily with 6-1/2 in. (Wray says his companion is "very difficult with 7 in. refractor, really only a glimpse star, even with the best atmosphere"), and have added two more stars to the group.

β *Delphini*.—Close double discovered by Burnham, est. 0".7. In contact 6-1/2 in., power 430.

η *Coronæ Borealis*.—Very low, long past meridian, in contact 164 power.

β *Equalei*.—[a 5-1/2, b 13, c 14, n 16: a-b 35", a-c 50" b-B 3".] 16^m. not seen, Webb 9-1/3 in. Just divided 164, easy 430 power. Two other companions (not seen by Smyth or Webb?)

μ *Andromedæ*.—The 16^m. comes, a very difficult test, easy; other comites (not seen by Smyth or Webb?) seen.

Companion to Vega pretty easy, Nov. 12th, 37m. after sunset.

P. 178 xx. *Delphini*.—[a 7-1/2, b 8, c 16, B 9: a-b 14".3, b-B 0".7, a-c 20".] Smyth 16 by evanescent glimpses; easy 164 power, 6-1/2 in. 8^m. well elongated in direction of 230°, power 430.

From H. BLYTH, Esq.

REGENT ROAD, GREAT YARMOUTH,

October 24th, 1874.

DEAR SIR,—Doubtless you have anticipated hearing from me, relative to the 10 inch telescope, but the weather has been so indifferent for delicate astronomical work, that it has not yet had a fair trial.

In middling good air, when I could use a reduced aperture with satisfactory results, the star images reminded me of the 6-1/2 inch, which I considered so superb a mirror that nothing could surpass its fine performance either on stars or planets.

With the recollections of the truly splendid views I had of Jupiter last season, I count much to see him with the 10 inch.

Yours truly,
(Signed.)

From T. AYERS, Esq.

REGENT ROAD, GREAT YARMOUTH,

February 2nd, 1875.

DEAR SIR,—In reply to yours, I have not had a really good night to test severely the defining powers of the 10 inch mirror, but judging from the few difficult objects I have seen, the mirror promises well, and I have no doubt of its excellence.

I had a fine view of the grand nebula in Orionis, and noticed that the 5th and 6th stars were quite plainly seen, when the aperture was reduced to 5-1/2 inches. I consider your stand a great improvement, being very convenient and steady.

Yours truly,
(Signed.)

From THE REV. A. P. ARNOTT.

AM STREET, EDINBURGH,

8-1/2" ALT. AZIMUTH.

February 25th, 1878.

DEAR SIR,—I have found no difficulty in putting up the instrument, and the adjustments do not appear to be hard to manage. It is a very beautiful instrument, and the case with which its movements are made is extraordinary.

I am, dear Sir,
Yours faithfully,
(Signed.)

Mr. G. CALVER.

From J. SLATER, Esq.

TOWN HALL SQUARE, BOLTON,

September, 1877.

DEAR SIR,—Last night, through a break in the clouds, I managed to get my first look with the telescope (an 8-1/2"). I turned it on delta Cygni. The definition was rather fluttering, but a 6-1/2 in. stop enabled me to see it well with 170; the first time I have ever divided this star^[2] the colour was obvious. I then tried Pi Aquilæ, which was beautifully divided. I am much pleased with the instrument and its performance, and when I do get a fine night I expect great things from my 8-1/2 in.

Yours truly,
(Signed.)

Mr. G. CALVER.

Footnote:

[2] The observer had used an 8-1/2" reflector, however, for some years.

(Another Letter from the same Gentleman.)

BOLTON,

December, 1877.

DEAR SIR,—As to the adjustments of the 8-1/2 in., I have had very little trouble, and I have tried it again on a few objects, a few nights ago, and with a 7 in. stop, and a power of 300, I split Zeta Cancri and the close double *sf* Castor; with the full aperture they were very easy. I have also split Gamma Andromedæ with 7 in. stop, and on a very fine night. I believe 6-1/2 in. would do it.

I just got a look, on November 7th, at the shadow of Titan, on Saturn's disc; but was driven inside before I could observe the time of its passing the centre. I have seen the Moon on two occasions with the finest definition I ever saw. Altogether I am highly pleased and proud of my telescope. I have not yet tried many faint objects; still I saw the Crab nebula in Taurus much nearer Lord Rosse's drawings than Herschel's, as given by Chambers.

I am, dear Sir,

Yours truly,

(Signed.)

Mr. G. CALVER.

From G. BILLINGTON, Esq.

WELLWICK, WENDOVER,

July 19th.

DEAR SIR,—I am perfectly satisfied with the mirror, a 6-1/2 in., now that I have tried it with *your* flat, and I have no difficulty in seeing star disks when the air is good.

I am, dear Sir,

Yours truly,

(Signed.)

Mr. G. CALVER.

From THE REV. S. TOMLINSON.

CARLISLE TERRACE, MALAHIDE, 1877.

DEAR SIR,—I write to tell you that I was very much pleased indeed with last night's observations with your 18". The star disks were good and well condensed, and stray light well gathered in, and when a good night comes I have no doubt of it.

The markings on Mars were well defined. I find the advantage of a large aperture shows itself. I have no doubt I picked up the outer Satellite, but cannot be certain till to-night, if I am permitted to see it. The only objection I have is its short focus, as it seriously complicates my labours in matching the secondary convexes for the Cassegrain form—if you can help me in the matter of a longer focus; if not I am satisfied and thankful, and must only proceed to the testing process inch by inch.

Yours truly,

(Signed.)

Mr. G. CALVER.

From the same Gentleman.

CARLISLE TERRACE, MALAHIDE,

November 6th, 1877.

MY DEAR SIR,—I had intended to write to you before, but, from various duties, I have been obliged to postpone it from day to day. I have been working steadily, however, with your 18", and I like it the more I use it. The

figure is wonderfully perfect, and the definition of planetary details and markings most admirable.

On four occasions I have seen the Satellites of Mars; of these I can feel quite certain of two, for it was held sufficiently long to perceive a marked change in position angle. The intensely small companion to α^2 Capricorni I have well seen, cleanly separated, and that in strong daylight. I am anxious for a good night to attack τ Orionis. During these observations I have been using the telescope as a Newtonian during my work at the convexes; but my labours have been pleasantly intercepted by the magnificent and gorgeous views of the star clusters and nebulæ—more beautiful and startling than I could possibly have imagined, and surpassing far anything I had ever pictured to my mind.

Yours very truly,
(Signed.)

Mr. G. CALVER.

From MAJOR-GENERAL ADDISONS, C.B.

MELTON, 1878.

SIR,—You will be glad, I know, to hear that I had a very satisfactory evening with the 18 in. equatorial yesterday, and the clock kept it in the field without vibration of any sort.

Yours obediently,
(Signed.)

Mr. G. CALVER.

From J. T. STEVENSON, Esq.

AUCKLAND, NEW ZEALAND, 1877.

DEAR MR. CALVER,—Since writing to you I have been observing with the 6-1/2 in. equatorial telescope, and I have much pleasure in informing you that I am greatly pleased with its performance. The definition of the planets is beautiful.

I am happy to inform you that I have found a most minute crater in Picard *a*, it is much smaller than the test crater (mentioned in Cel. Obj., by WEBB, page 83) in same formation. I wrote to Mr. WEBB about it, and since then I have seen it distinctly; it is another proof of the excellence of your telescope.

Yours truly,
(Signed.)

Mr. G. CALVER.

From J. C. HARKER, Esq.

KENDAL, 1878.

DEAR SIR,—After four months use of the telescope I had of you (an 8-1/2 in.), I am able to tell you that I am much pleased with its great excellence. The definition of the mirror is very fine, and its performance on the Moon being superb, with so high a power as 700. The stand I find very steady and convenient, and the screw motions very smooth.

I am, dear Sir,
Yours truly,
(Signed.)

Mr. G. CALVER.

From E. A. FRY, Esq.

BIRMINGHAM, 1878.

DEAR SIR,—I write to tell you that I like the telescope very much (5-1/4 in. equatorial); the mirror is, as you say, a gem, and it is quite a treat to look at the mysteries above. Saturday was a glorious night, though the Moon was rather too bright. Altogether I am quite satisfied with the telescope, and the finish is first-rate.

Yours truly,
(Signed.)

Mr. G. CALVER.

From THE REV. J. M. COATES.

THE GODDARDS, MOULTON, SPALDING.

DEAR SIR,—I am able to report satisfactorily on the mirror. On Saturday night last I divided μ^2 Boötis, power 300, with the 8-1/2".

Yours truly,
(Signed.)

Mr. G. CALVER.

From the same Gentleman.

THE GODDARDS, MOULTON, SPALDING,

January 18th, 1879.

DEAR SIR,—I have been waiting for opportunities for testing observations, but there have been few nights when the weather and leisure time have concurred in giving me a chance. However, I can say this much, that in lunar work I have had much pleasure from the excellent definition given by the 8-1/2 in. I had some fine views of Copernicus under high powers not long since, and am anxiously awaiting opportunities for closer study of a particular region.

Yours truly,
(Signed.)

Mr. G. CALVER.

From FRANK C. DENNETT, Esq.

10, TERMINUS TERRACE, SOUTHAMPTON,

December 10th, 1878.

DEAR SIR,—The 5-1/4 in. is beautiful. Last night there was a slight frosty fog, and the air was very very steady, in fact I only remember about one such night since November, 1875. Saturn was sharp and hard, and the shadow of the ring intensely black, with all powers from 168 to 372, and even with 550, obtained by removing the field lens of 372. The very broad polar caps presented a bluish grey, rendered more visible by contrast with a fine penumbral line of a very slightly brownish grey close to the south border of the ring, and which seemed to be diffused on the south border (*i.e.*, the line was diffused). I should fancy the line here referred to was less than 1" of arc broad; considerably less!

Turning from Saturn, I looked at ϵ Arietis, the present distance of which is only 1"·17; with all powers, from 213 upwards, the components were visible as two tiny points of light thoroughly divorced, and with 288 power, reminding one of ϵ^2 Lyræ, as seen with 120 on a 2-3/4 in. achr., only the discs seemed freer from rings.

Later in the evening, when the definition was more fluttery, η Orionis shared the same fate as ϵ Arietis, being well separated with the same powers. The present distance of this star is only 1"·12. Recently I divided this star with a 4-1/2 in. stop on the mirror, power 288. I have seen ϵ Arietis well with the same stop too. I think I have told you before that the 5-1/4 in., with 372 power, was successful in dividing ϵ Equulei, whose distance is only 0"·97. This is the closest star I have yet divided.

The Moon is fine too! Last night I caught a large ring close to the terminator, near the south pole (for the Moon was only about 2-1/2 hours

past full), which was looked at over the northern wall; the view was beautiful, such gradations of light and shade, interspersed with greys of different hues, which render the object scarcely describable. The finest clefts and tiniest craterlets are rendered visible. It is a puzzle to me how anyone in the possession of £16 or £17 can go buying a refractor of from 2-1/2 to 3-1/4 in. aperture in preference to a silvered mirror. It must be from sheer ignorance of the beautiful views given by the latter, which is superior in every way.

I do not think there is any other point on which I need touch now, so I must beg to remain,

Dear Sir,
Yours very truly,
(Signed.)

Mr. G. CALVER.

From THE REV. JEVON J. MUSCHAMP PERRY, M.A., F.R.A.S.

ST. PAUL'S VICARAGE, ALNWICK,

December 20th, 1879.

MY DEAR SIR,—I was very much interested in reading your account of the grinding and polishing of the 37" speculum, in this week's E. M., copied from the "Monthly Notices" of the R. A. S. Had I only the wealth of my neighbours, I would give you an order for the largest telescope you could construct. Every really good night gives me some fresh proof of the great excellence of the 6-1/2 in. For instance, on December 15th, I clearly and steadily saw *Enceladus*. I could not, of course, have recognized it had it not been that this month's "Observatory" gave diagrams of the positions of the five inner sats. of Saturn, for 8 p.m. on every day of this month, and a careful search for Encel. was fully rewarded by a clear and steady view of it at that time in its proper place. On this feat the Rev. T. W. WEBB warmly congratulated me, and said that he had seen Encel. with his 9-1/3" "With," but only when Saturn was hidden behind a bar. I saw it clearly in exactly its

proper position *without* hiding the planet—power 400. I have also had several good views of Mr. WEBB'S new nebula in Cygnus, and have sent him drawings of the field, which he confirms. I have also made some fine drawings of Jupiter and Mars. On December 12th, 11 p.m., I tackled σ 749 (in Taurus), a very difficult test for my aperture, only 0·8", and *perfectly* split it with power 300. The night was superb. When you consider that the theoretical limit of dividing power for a 6-1/2" is only 0·7", I think I have done well to split a star 0·8" in our climate.

Faithfully yours,

(Signed.)

Mr. G. CALVER.

From G. FERGUSON, Esq., M.D.

ALTIDORE VILLA, PITTVILLE, CHELTENHAM,

June 26th, 1879.

MY DEAR SIR,—I write to tell you that Thursday night being very fine, I had a good opportunity of trying the (an 8-1/2") mirror on Jupiter, although this planet was lower than it should have been to give the fairest trial. Well, I can only assure you that, despite the low position, I never saw it so beautifully before. The contour of the clouds, particularly one orange one, *sf*, the delicate lines, and the colouring of the belts, were all seen with a sharpness which I had believed scarcely attainable, and proving that the result of your labours has been to effect a very real improvement in the speculum. Notwithstanding the good planetary definition, the air was not tranquil enough for seriously attacking really difficult doubles; still I was much pleased with its performance on π Aquilæ.

Yours truly,

(Signed.)

Mr. G. CALVER.

From the same Gentleman.

July 27th.

Yours of July 25th duly to hand. I have the speculum adjusted in the “pink” of accuracy. Star discs are small and round, with concentric diffraction rings, and quite rayless except in the case of the very brightest stars, and even with these the rays are very trifling. But for the diffraction rings, which I fancy a really good night will mostly get rid of, I should, I feel sure, have readily divided η Coronæ.

The Moon last night, though very low, was seen with most gratifying ease and sharpness. Altogether you will gather that I am more than satisfied. I am greatly gratified, and will look forward to great pleasure as the season advances.

Yours faithfully,

(Signed.)

Mr. G. CALVER.

From THE REV. JEVON J. MUSCHAMP PERRY, M.A., F.R.A.S.

ST. PAUL'S VICARAGE, ALNWICK,

August 8th, 1880.

MY DEAR SIR,—You have indeed surpassed yourself, and your performance is better than your promise. The mirror arrived safely yesterday, and when I unpacked it I was exceedingly pleased with its surface and general appearance, but of course that was a small matter. I waited till night to test it on a close double.

I first tried it on ϵ Boötis, and although the night was very bad the definition it gave of the two stars with 450 and full aperture was admirable, considering the night. After waiting a long time for clouds to clear away, I turned the tube on that exceedingly difficult test, δ Cygni. I consider this star to be one of the severest tests for any telescope, reflector, or refractor of any aperture in the whole heavens. The performance of the mirror upon it,

with the *full* aperture, was simply MAGNIFICENT, the small close companion standing out clearly, well separated from its bright primary, although the night was not good. I hardly thought it possible that the fine 18 inch mirror could be excelled. I am bound to confess that you have excelled it.

To say that I am extremely grateful to you is a very poor expression of my feelings. I am pleased and delighted *beyond measure* with the *super-excellent* 18-1/4 in. mirror you have sent me. Clouds put an end to my work last night, but I have no doubt that in the future every night's observations will only accumulate proofs of the exquisite perfection of your handiwork.

Very faithfully yours,

(Signed.)

Mr. G. CALVER.

Extract from the "English Mechanic," September 3rd, 1880.

"The telescope is a silvered-glass Newtonian reflector. There are two specula, one of 18 in. in diameter, figured by the celebrated WITH, in 1878, and is a magnificent specimen of that eminent maker's handiwork. In fact, I am given to understand that Mr. COOPER KEY considered that he had attained perfection in this mirror, and of all men then living there was no better judge than he. The other is 18-1/4 in. in diameter, and was originally figured by Mr. COOPER KEY himself; but it was only perfect over the interior 14 in., and, consequently, while the whole aperture could be used on the moon and planets, this was all of it that could be used upon close doubles. I determined, therefore, to give this mirror to Mr. CALVER to re-figure, and, if possible, excel Mr. WITH's speculum.

"In order that Mr. CALVER might know the extent of his task, I returned home through London and Chelmsford, bringing with me both mirrors, which Mr. CALVER tested for himself at the centre of curvature. After the trial was over, he pronounced Mr. WITH's mirror to be an exceptionally fine one, and heartily congratulated me on the possession of such a beauty. I left the other one with him to be re-figured.

“And now to come to the most important part of all—the performance of the 18 in. WITH mirror upon close double stars. Since the telescope came into my possession, I have never had a thoroughly good night; but even on an indifferent one, I have, with the *full aperture*, divided such extremely difficult tests as η Coronæ, μ^2 Herculis, and δ Cygni; and these results were quite sufficient to show me that the very high reputation of its excellent maker is worthily and proudly maintained by this mirror. In fact, the definition it gave one night, when the Scottish Astronomer Royal was staying with me of ϵ Boötis, was such, that the Professor told me I could not expect any telescope, reflector, or refractor, upon the same object, and on the same night, to do better. However, I knew Mr. CALVER was doing his very utmost in the honourable and praiseworthy competition to attain perfection, to give me a mirror which would bear favourable comparison with Mr. WITH’s. It reached me about a week ago, and on the same evening I had an opportunity of testing it upon what I consider to be one of the very severest tests that can be applied to any telescope, reflector or refractor, in the northern heavens, that is δ Cygni^[3]. To make the test as severe as possible, I applied the *whole* aperture at once, with a power of 347. I was surprised and delighted *beyond measure* with the result. The small star stood out clearly separated from its bright primary, and there were two or three diffraction rings round the latter *unbroken* in any part, and *equally* bright all round. I consider the results which Mr. CALVER has attained in this mirror cannot possibly be excelled. I measured this star on two separate nights, and the mean results are pos. 328.9° , dist. $1.29''$. I also measured, with this mirror, another difficult star of the same character, namely, with a bright primary and small companion, ζ Herculis, with the following results, pos. 111.3° , dist. $1.03''$. I make the companion of this star a dusky blue, or lavender. As regards the light-grasping power of the mirror, I turned it, the other night, on the minute pair between β^1 and β^2 Capricorni, which Sir John Herschel gave as the test of the power of any telescope and eye to observe the satellites of Uranus. The night was very unsteady, but the minute stars of the pair were most conspicuous.”

Footnote:

[3] NOTE.—Mr. Perry does not mean that δ Cygni is one of the most difficult tests as regards the distance of the components. He

has often seen it with his 6-1/2" Silvered Glass Reflector, and even with a stop of 5-1/2" on the same instrument; but he was surprised at the beauty and perfection of the images—perfect and unbroken rings—and he knew that these appearances could not have presented themselves had not the *figure* of the mirror been of extraordinary perfection.

From THOS. WESTLAKE, Esq.

FORDINGBRIDGE,

June, 1880.

MY DEAR SIR,—I wrote you last evening, and afterwards had my first trial with the telescope (a 10 in. Alt Azimuth). It turns out a beauty. I had the full aperture on all the evening, the air was very good, but not perfect. Alpha Lyræ was a splendid object with all powers, a most brilliant disk with just indications of circular rings—no wings or rays. I turned it on B Cygni, and the beauty of the mirror stood revealed, both stars *perfect disks*, with *just* the indication of perfect circular defraction rings, and the colours were most beautiful,—nothing could possibly appear finer. I consider the mirror perfect, and the stand most convenient and smooth and pleasant in action.

I am, dear Sir,

Yours truly,

(Signed.)

Mr. G. CALVER.

From THE REV. E. H. LOWE.

WILMCOTE,

April, 1880.

DEAR Mr. CALVER,—I have been exceedingly occupied, as there has been a succession of gloriously fine nights, and I have secured the Nebulæ diagrams I so much wished for Mr. WEBB,—and I would allow nothing to divert me from them—but I put it (a 15-1/2 in. equatorial) on Zeta Cancri, and it was shown beautifully. I have been working at the driving clock, and it is under capital control. The great cluster 13 M., is a glorious object, and magnificently resolved. I see all I look for on the Moon.

Faithfully yours,
(Signed.)

Mr. G. CALVER.

From W. SCOTT, Esq.

SOUTHWICK, SUNDERLAND,

July, 1880.

DEAR SIR,—Dr. Blacklock, and also Mr. Bell, have seen the mirror, and pronounce it perfect, and it performs admirably.

Yours truly,
(Signed.)

Mr. G. CALVER.

From JAMES L. McCANCE, Esq., F.R.A.S.

PUTNEY HILL, S.W.,

October 23rd, 1879.

DEAR Mr. CALVER,—Weather and circumstances have been very unfavourable for doing much work with the 10 inch reflector since I got it from you. But with what I have seen with it I am very pleased; in good

weather the views of the planets, Jupiter and his “Great Red Spot” especially, were grand. The mounting (equatorial) is quite satisfactory, and the movements work smoothly and well.

Yours very truly,

(Signed.)

From FRANK C. DENNETT, Esq.

10, TERMINUS TERRACE, SOUTHAMPTON,

July 11th, 1880.

DEAR SIR,—It is a long time ago that you asked me to write and let you know how I liked the 9-1/2 in. I have not done so because I wished first to see what was to be made of the planets.

Last night, at 14^h 30^m, I had my first peep at Jupiter through it, and a grand sight it was; his belts coloured, though not so vivid as they are sometimes, were fine in the extreme; and, though the power employed was only the achromatic 215, the satellites were beautiful little discs, one especially showing the tint of its surface.

A quarter of an hour later Saturn was in the field of view; a grand object, several of the satellites surrounding it. A little cloud was coming up over it, so I could not see so much of this planet as I should have liked, though amply sufficient to prove incontestably its high class.

With double stars I have not attempted much, though on turning it on μ^2 Boötis I found the division easy, the components being respectively slightly orange, and dullish white or pale yellow.

I have had lots of grand views of the wonders of the “star depths,” clusters, and nebulae, notably 57 Messier, the Ring nebula, in Lyra, the great Orion nebula, and the marvellous Whirlpool nebula, the chief outlines of which were well shown.

Hitherto the instrument has been chiefly employed on the Moon, the details of which come out hard and sharp in a beautiful manner, with all powers up

to 515, the highest I have.

The 5-1/4 in. pleased me very much, but the 9-1/2 in. delights me.

Yours faithfully,
(Signed.)

Mr. G. CALVER.

From G. W. ROYSTON PIGOTT, Esq., M.D., F.R.S., F.R.A.S., &c.

EASTBOURNE,

July, 1880.

DEAR SIR,—There has been but little weather for testing. Mr. Slack has been staying with me; we are delighted with the 8-1/2", and think it a great beauty. I can hardly imagine any mirror much better, and I expect great precision from it. I find its focus to be 65 inches, not quite so much as my fine Dollond glass, of 3-3/4 in. aperture.

Yours truly,
(Signed.)

Mr. G. CALVER.

From A. WILLIAMS, Esq.

CLARENCE LODGE, GRAFTON STREET, BRIGHTON,

August, 1880.

DEAR SIR,—I had intended writing to you to express my pleasure in my 5-1/4 in. telescope nearly a month ago, but thought I would have another look first, and the more I look the more I am pleased with it. I cannot now imagine anyone preferring a 3-1/4 in. refractor to one of these. I have had no trouble at all with the adjustments, and if was to get out of adjustment, it

could be rectified in about five minutes. Notwithstanding the unsettled weather (giving on an average only about one fine night in eight), I have examined nearly two hundred double stars in the last three months, some of them very difficult. I enclose a list of a few, which will show very well the capabilities of the instrument.

Yours truly,
(Signed.)

Mr. G. CALVER.

LIST OF OBJECTS SEEN WITH A 5-1/4 INCH REFLECTOR.

ξ Scorpii (5, 5.2 : 1".3) clearly divided, 186 power.

ϵ Lyræ. Debilissima quite easy; 186, 380 powers, though nearly half Moon.

μ^1 Herculis (9.4, 10 : 1".04). Just divided, 380. Very difficult, owing to its faintness.

Vega. 11 mag., comes quite plain, only 47 minutes before sunrise; powers 110, 186.

ϕ Virginis (5, 13 : 4".27). Comes plain, 110, 186.

ν Scorpii, the close pair (4, 5 : 0".8), well elongated, power 380; small star (7, 8 : 1".68) divided, 110.

ζ Boötis (4, 5 : 0".88) elongated 186, notched 380, and with field lens removed, all but divided.

δ Cygni (3, 9 : 1".63), 9 plain, 380 on several occasions.

Antares is chiefly an atmospheric test, but the companion was quite plain on one occasion out of the three I looked for it.

μ^2 Boötis (7.5, 8 : 0".63), considerably elongated, 380.

η Coronæ (5.5, 6 : 0".62), slightly elongated, 380; but not nearly so much as μ^2 Boötis.

56 Herculis (6.5, 13 : 15"), 13 easy, all powers.

π Aquilæ (6, 7 : 1".5). Just separated, 110; 14 mag. (Webb), just visible, 186.

ε Equulei. Just divided, 186 (5·7, 6·2 : 1"·0).

From D. TRAILL, Esq.

BALHAM,

March 5th, 1880.

MY DEAR SIR,—I have at present to use the telescope, an 8-1/2 in. Alt Azimuth, out of a window, and so at a great disadvantage,—in fact cannot give it a fair trial till I have the house ready for it, meanwhile I use it as occasions offer, and have been highly pleased with it. The views of the nebular in Orion resemble those of BOND's drawings, and shows immeasurably above my 3-1/4, of course. I think the mirror is *perfect* from its performance on large stars whose disks are small and round, without rays of any kind. I am *thoroughly* satisfied.

Yours truly,

(Signed.)

Mr. G. CALVER.

From HUGH AULD, Esq.

GROSVENOR CRESCENT, EDINBURGH, 1879.

DEAR SIR,—I have not been able to do much with the telescope, a 5-1/4" Angle-block, through the unfavourable weather for the last four months, but with the opportunities I have had its performance is satisfactory; it performs well on stars with high powers, and I am sure it will bear a good deal higher. I am very well pleased with the instrument.

Yours truly,

(Signed.)

Mr. G. CALVER.

From THE REV. THOS. H. BROWNE, F.R.M.S., &c.

THE CEDARS, HIGH WYCOMBE,

February 26th, 1880.

DEAR SIR,—I am exceedingly pleased with the action of your mirror (an 8-1/2 in. equatorial); what I have seen of sun, moon, planets, and stars is splendid. I have had a good many telescopes, but never liked one so well as the one you have sent me. For the first day or two I could do nothing but admire it.

Yours truly,

(Signed.)

Mr. G. CALVER.

From THE REV. CONYBEARE W. BRUCE, M.A., F.R.A.S.

ST. NICHOLAS' RECTORY, CARDIFF,

July 17th, 1880.

MY DEAR Mr. CALVER,—You will have been looking for some news of me and the 12". I have purposely delayed writing until I was *really* in a position to say something definite, and you had already heard my high opinion of the mirror. Let me at once say that that opinion has been *more* than justified. The mirror is a *beauty*. Canon Beechey and I got Jupiter and Saturn at 2.30 a.m. yesterday morning—and they were *superb*. The Canon declared enthusiastically, that he “had *never* seen Saturn anything like that,” and the last thing he said to me on the subject as I parted with him to-day was, “You have got a very fine instrument indeed.”

If I *could* add anything more in its praise it would be this, that I am really going to build a *new observatory* for it—my present one is too small and cramped.

Sincerely yours,

(Signed.)

Mr. G. CALVER.

(Another Letter from the same Gentleman.)

ST. NICHOLAS' RECTORY, NEAR CARDIFF,

August 28th, 1880.

DEAR Mr. CALVER,—The 12 inch surpasses all my expectations. It has the perfection in figure of the incomparable 10 inch of yours, which was its immediate predecessor, and, in grasp of light, the difference in favour of the 12 inch over the 10 inch, a most brilliant mirror, is simply astounding.

You may equal the 12 inch—I am sure you will, whenever you make one—but you will not easily beat it. Mr. SADLER, the well-known and most accurate observer, has himself worked with it, and writes of it to me as follows:—"The 12 in. is considerably superior in light grasping power to H₂'s 20 ft. reflector, and the figure is a much finer one."

Your skill and kindness has placed me in the front rank among amateur observers, as far as the possession of optical means is concerned.

I was afraid that the German Stand would not prove so convenient as the "Berthon Equestrian Stand," but I find it is and far more so. The German Stand is, for anything above 6 in., decidedly more solid and dependable, and far simpler in any case. The merit of the "Berthon Equestrian" form is that it requires a smaller observatory.

Very sincerely yours,

(Signed.)

From W. S. FRANKS, Esq., F.R.A.S.

LEICESTER,

August 19th, 1880.

MY DEAR SIR,—As the 11-1/4 in. speculum has now been in constant use for two years, I may be allowed to express my opinion, with all due deliberation, as to its performance. It has been tried upon almost every conceivable object during that period with uniformly good results. Having also a 5 in. achromatic of the finest quality, I have been enabled to make a great number of comparisons between the two instruments, mounted side by side: and for fine definition, smallness of star discs, and freedom from chromatism, the reflector always came out victorious. Upon first mag. stars, nothing could surpass the purity of the image in the reflector; but on immediately turning to the achromatic, the outstanding violet rays were painfully obtrusive. For red stars, which are generally small, the reflector seems especially adapted. I used to be rather sceptical about the tints of red stars, but must confess, after seeing the magnificent tints of scarlet and crimson which are shown by a large aperture, that my doubts have vanished. Definition upon the planets, in good air, is simply perfect. The detail on Jupiter and Saturn reminds one of an exquisite engraving. The far distant Neptune is perfectly defined as a tiny disc, and his moon steadily visible with the 11-1/4 in. The faculæ and other Solar detail are brought out with wonderful distinctness. As for the Moon, the crispness of definition, absence of colour, and immense number of minute objects shown by the reflector, leave nothing to be desired. The gorgeous views of clusters and nebulæ: the brilliantly crowded fields in the galaxy: and the endless variety of star colours and groups are almost beyond description, and must be seen to be appreciated. I could fill pages about objects seen with the 11-1/4 in. mirror, but will not occupy more space, save to thank you for having placed such perfect instruments within the reach of amateurs.

Yours faithfully,

(Signed.)

Mr. G. CALVER.

From W. S. FRANKS, Esq., F.R.A.S.

LEICESTER,

September, 1878.

MY DEAR SIR,—Mr. BURNHAM is undoubtedly “facile princeps,” as an observer of double stars. Few men could have discovered such difficult objects as he has done with a 6" O.G. Perhaps, it ought to be conceded that the usual formula for separating power, requires modification in his case, as Pogson’s formula for “minima visible” did with Mr. J. W. WARD, of Belfast. Apropos of this topic: do you recollect Mr. WARD’S discovery of several additional comites to Pxx 452 Cygni? One of them Mr. BURNHAM only succeeded in seeing with the greatest difficulty with the 6" O.G. when the large star was hidden by a bar. I can see this easily enough with the 11-1/4 in. mirror, in presence of the brighter star.

When you have any intending customers halting between two opinions, or in any way dubious about reflectors—if you will refer them to me, I shall have great pleasure in giving my testimony. Reflectors have been very badly used, and in justice to them I shall not hesitate to express my honest conviction whenever it is asked.

Yours sincerely,

(Signed.)

Mr. G. CALVER.

From PHILIP VALLANCE, Esq.

COBB COURT,

August 24th, 1880.

DEAR Mr. CALVER,—I write to thank you for the very excellent 12 in. mirror you have worked for me. You know that I have excellent opportunities of testing it by those of With, Bird, Grubb, Murrell, and others; and also for definition by a capital 6 in. refractor—and I can truly say it is surpassed by none, and equalled by few: it is quite my favourite instrument, and has been

a source of much enjoyment ever since I have had it, and again I thank you for working me such a treasure.

I am, dear Mr. Calver,
Yours very truly,
(Signed.)

From THE REV. N. S. GODFREY, F.R.A.S.

ST. BARTHOLOMEW'S VICARAGE, SOUTHSEA,

August 23rd, 1879.

MY DEAR Mr. CALVER,—The 12 inch mirror which you worked and mounted for me gives me perfect satisfaction. I think I have had sufficient experience with telescopes to justify the expression of a confident opinion, as at one time or other I think I have had 13 achromatic telescopes by Tulley, Dollond, Simms, Jones, Cooke, and Murrell, of apertures varying from 2-1/4 to 7 inches, besides 5 graduated instruments. The one I had before yours was an 8-1/8 With-Browning, which With himself considered of as fine a figure as he could work, and it certainly was a beautiful mirror. Your 12 inch is quite as perfect in its definition. I have soon on particularly fine evenings Castor as two discs, without wing or ray. I have separated with ease ω Leonis, μ^2 Boötis, η Coronæ, γ^2 Andromedæ, and many others of that class. The planets Jupiter and Saturn on fine nights are magnificent. I could go into more detail, but I think the facts I have mentioned are sufficient to convey an idea of the performance of the mirror; and I am particular in expressing my opinion thus strongly, because great names, with no experience of reflectors, are committed to their condemnation. I am certain that under favourable circumstances they are *quite* equal in their performances to refractors. I know my 8-1/8 With was in every way equal to my 7 inch Troughton and Simms, and in light superior, when newly silvered. And when the cost is taken into consideration, Cooke's price of a 10 inch O.G. being £390, without mounting, and your price for a 10 inch mirror being only £25, I can only express my astonishment that there can exist any doubt in the scientific mind upon the subject.

May I say that I shall be pleased to answer any questions that any of your intending purchasers may wish to ask. I cannot speak too highly of your mirror. I may add that through the courtesy of its possessor, A. A. COMMON, Esq., I saw Jupiter magnificently last year in the 37" mirror you worked for him.

I am, my dear Mr. Calver,
Very truly yours,
(Signed.)

From THE REV. CANON ST. VINCENT BEECHEY, F.B.A.S., &c.

HILGAY RECTORY, DOWNHAM, NORFOLK,

June 22nd, 1880.

MY DEAR CALVER,—I really do not know how to thank you enough for all the pleasure you have given me, and for the excellence of the work you have done for me. I like it more and more! It is really as near perfection as I could ever hope for. Even at my old age of nearly eighty, I divide 1 second stars like two dots, and I used a power of 700 on my 8 foot. What mirror but of great excellence would bear that power, and give you two *dots*! I would not part with it for anything. When I have been to Mr. Bruce he is coming to me, and we will try your 12" against my 10". He will have light, a great thing for me, but I do think he *cannot* have *better* definition. I hope nothing will prevent his getting the 12" in time for us to adjust and try it. My telescope, now complete, is a gem, and has given me more pleasure than any other scientific thing.

Ever most sincerely yours,
(Signed.)

Mr. G. CALVER.

From THE REV. J. GWYNNE HUGHES, F.M.S., &c.

MALDON,

August 30th, 1880.

MY DEAR SIR,—All my friends are greatly pleased with the 10 in. reflector you have made me—with its general appearance and superior workmanship.

The wheel at the foot adds very much to its portability. Tho form of its stand is the best I have seen, giving a complete sweep without any fresh adjustment. The cell bearing the reflector drops into its place without any difficulty, even in the dark. The trials I have made of the instrument have revealed a clearness and sharpness of definition far superior to anything of the kind I had seen before.

I am indebted to you for a source of boundless scientific interest at very moderate expense. With best wishes,

I remain, dear Sir,

Yours sincerely,

(Signed.)

Mr. G. CALVER.

From WALTER GOODACRE, Esq.

SEVEN SISTERS ROAD,

May, 1879.

DEAR SIR,—Respecting the 5-1/4 in., I am very much pleased with it; the definition is very good.

As a proof of the excellence of the mirror, I may mention that I have on several occasions seen the small crater on the floor of Helicon, given by WEBB as a test for a 9 in. mirror.

Yours truly,

(Signed.)

Mr. G. CALVER.

(Another Letter from the same Gentleman.)

From REV. N. S. GODFREY, F.R.A.S.

October 4th, 1880.

DEAR SIR,—I have been able to get to use my telescope again, and send you some of my seeings.

June 17th.— α Herculis. Still night, 10 p.m., light haze. Discs beautifully small. Occasionally without any interference ring.

July 18th.— ζ Herculis. Beautifully seen, power 496, Cooke.

August 16th.—Well seen. 6.15 p.m., daylight.

August 26th.— σ Cygni. Misty. Adjustment beautiful. Small star well seen. ψ Cassiopeæ. Double blue star well divided.

August 27th.— ζ Herculis. Both stars occasionally without ray or ring. Two brilliant dots. Colour of companion, pale lilac. Full aperture. Power 207, Cooke. 6 p.m.

August 30th.—Dumb bell nebula. Well seen. Not at all regular in outline, as given by H. Several glimpse stars.

September 24th.— ζ Herculis. Very fine definition. The difficulty with this star is clearly atmospheric. I saw it well with 8 in. aperture, power 207: but the interference ring became thicker and continuous all round: probably owing to the larger size of my diagonal prism, nearly 2-1/2 in.

The planets are now again simply magnificent. All the usual details I see splendidly with full aperture of 12 in., and as sharp as if cut out with a penknife. The great red spot is a wonderful phenomenon. What is it?

The great nebula in Andromeda, seen through your fine nebula power is a mysterious sight. You seem to be looking at something which eludes your vision (except the nucleus); you see it, and yet, ghost-like, you see through

it. I long for the great nebula in Orion. I did not see it last year on account of illness.

Believe me,
My dear Mr. Calver,
Very truly yours,
(Signed.)

Mr. G. CALVER.

From T. HILDERSLEY, Esq.

33, BERWICK STREET, W.,

September 9th, 1880.

DEAR SIR,—I am very pleased to be able to tell you that I have got the 8-1/2 in. mirror into very fair adjustment—only a slight flare on one side, which I could correct if clouds had not come up, and I have not had a chance since, as my time is very limited.

As it was, comes to Wega came out fine; also debilissima, ϵ Boötis, μ^2 Boötis, was well seen. Stops made little improvement. Nebula in Andromeda very fine, both large and small. Clusters in Hercules well resolved, and Jupiter and Saturn, though low down, were grand; all is clear with any power or aperture.

Yours truly,
(Signed.)

Mr. G. CALVER.

From REV.E. A. FISHBOURNE, M.A.

MINOR CANONRIES, LLANDAFF,

October 4th, 1880.

MY DEAR SIR,—I send you my cheque, which I do with the greatest pleasure. I am delighted with, the telescope and with its capabilities.

Faithfully yours,

(Signed.)

Mr. G. CALVER.

From WALTER GOODACRE, Esq.

LONDON, N.,

September 6th, 1880.

MY DEAR SIR,—The 8-1/2 in. mirror which you supplied me with during the early part of the year, I have recently mounted as a simple equatorial. And although I have had but few opportunities at present of using the same, they have been sufficiently numerous to prove, beyond doubt, the great excellence of the mirror.

On the evening of the 3rd September, I had some beautiful views of Jupiter and Saturn. In the former the deep coppery hue of the equatorial belts, and of the large spot, at present visible on his southern hemisphere, were strikingly conspicuous, as were many minute details of his surface. In the case of Saturn, the gradations of light on the rings, and the delicately tinted belts on the planet itself, also the sharply defined shadow of the planet on the rings were easily seen.

Its definition of the most minute portions of the moon's surface, leaves nothing to be desired, whilst its performance on double stars is equally satisfactory.

Yours faithfully,

(Signed.)

Mr. G. CALVER.

From G. W. ROYSTON PIGOTT, Esq., M.D., F.R.S., F.R.A.S., &c.

EASTBOURNE,

October 4th, 1880.

DEAR SIR,—Your figuring of the 8-1/2 in. mirror has given me much satisfaction. I see $\epsilon^1 \epsilon^2$ Lyræ, and such stars, with 600, very sharp and clear.

I have counted six satellites of Saturn, and the equatorial belt is very distinct.

Last week I happened to observe Jupiter, when a black shadow of a satellite was beautifully notching the limb, at the same time a small pearl-like satellite was about to egress from the belts. I saw, also, six spherical cloudlets on the belts.

Yours faithfully,

(Signed.)

Mr. G. CALVER.

From REV. S. MILLS, M.A.

DROYLSDEN,

September 4th, 1880.

DEAR Mr. CALVER,—To-night was an exceptional fine night for the stars, the like of which I have not had since I got your 6-1/2 in. I had for a week or so been observing with my home-made 8 in., but to-night again recurred to *yours*, and its *superiority* was all the more *conspicuous* and *gratifying*. It quite *delighted me afresh*!

Looking at β Cygni, I picked up two, to me, new stars, which I think fully completes Mr. SADLER's tests in this field. I mark the newly found ones in red. This only shows that one has only patiently to watch and wait for a

suitable observing night to get one out of a certain aperture it is capable of showing.

The vast superiority of your 6-1/2 in. *has quite put my 8 inch out of court!!*

Yours faithfully,

(Signed.)

Mr. G. CALVER.

(From the same Gentleman.)

DROYLSDEN RECTORY, MANCHESTER,

July 30th, 1880.

MY DEAR SIR,—I can scarcely express to you the pleasure I have had in the use of your splendid 6-1/2 in. mirror. Both in light grasping power, and in the separation of close double stars, it has far surpassed my highest expectations, and quite removed the prejudice I entertained against reflectors. Though possessing a first-class 4-1/4 in. refractor, equatorially mounted in my observatory, I now use your mirror in preference, in the more difficult and delicate observations, and with it am able to master the most severe tests given for 6-1/2 in. aperture. I would not be without one of your finely figured mirrors, now that I have tested their merits, for any consideration.

(Signed.)

Mr. G. CALVER.

(From the same Gentleman.)

THE RECTORY, DROYLSDEN,

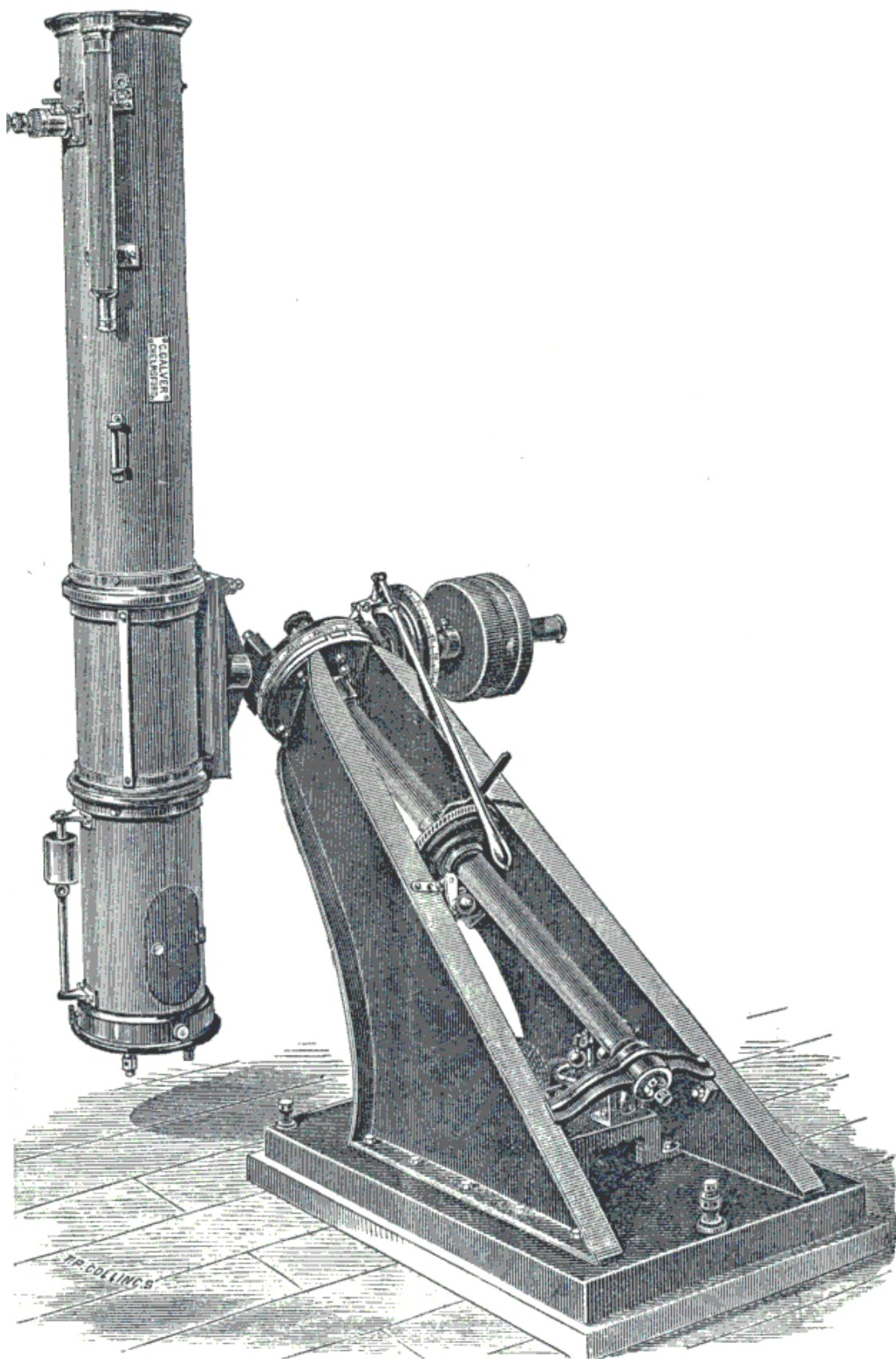
October 4th, 1880.

MY DEAR SIR,—I have given your 8-1/2 in. mirror one night's trial, and it has greatly pleased me. Its resolution of the great cluster in Hercules was truly magnificent, and the clearness with which it brought out stars very difficult with 6-1/2 in. was highly gratifying. I hope to turn it to very good account.

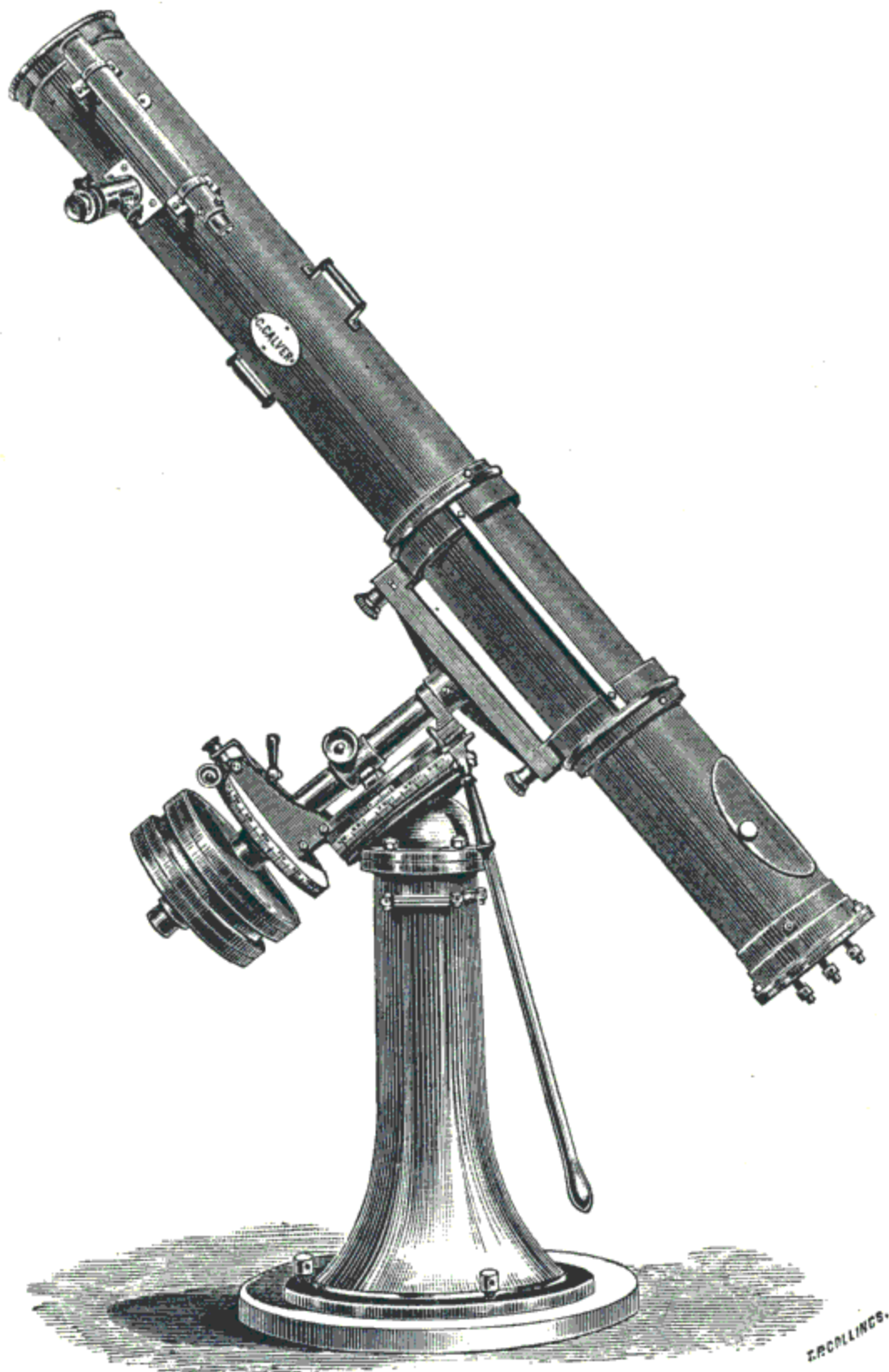
Yours faithfully,

(Signed.)

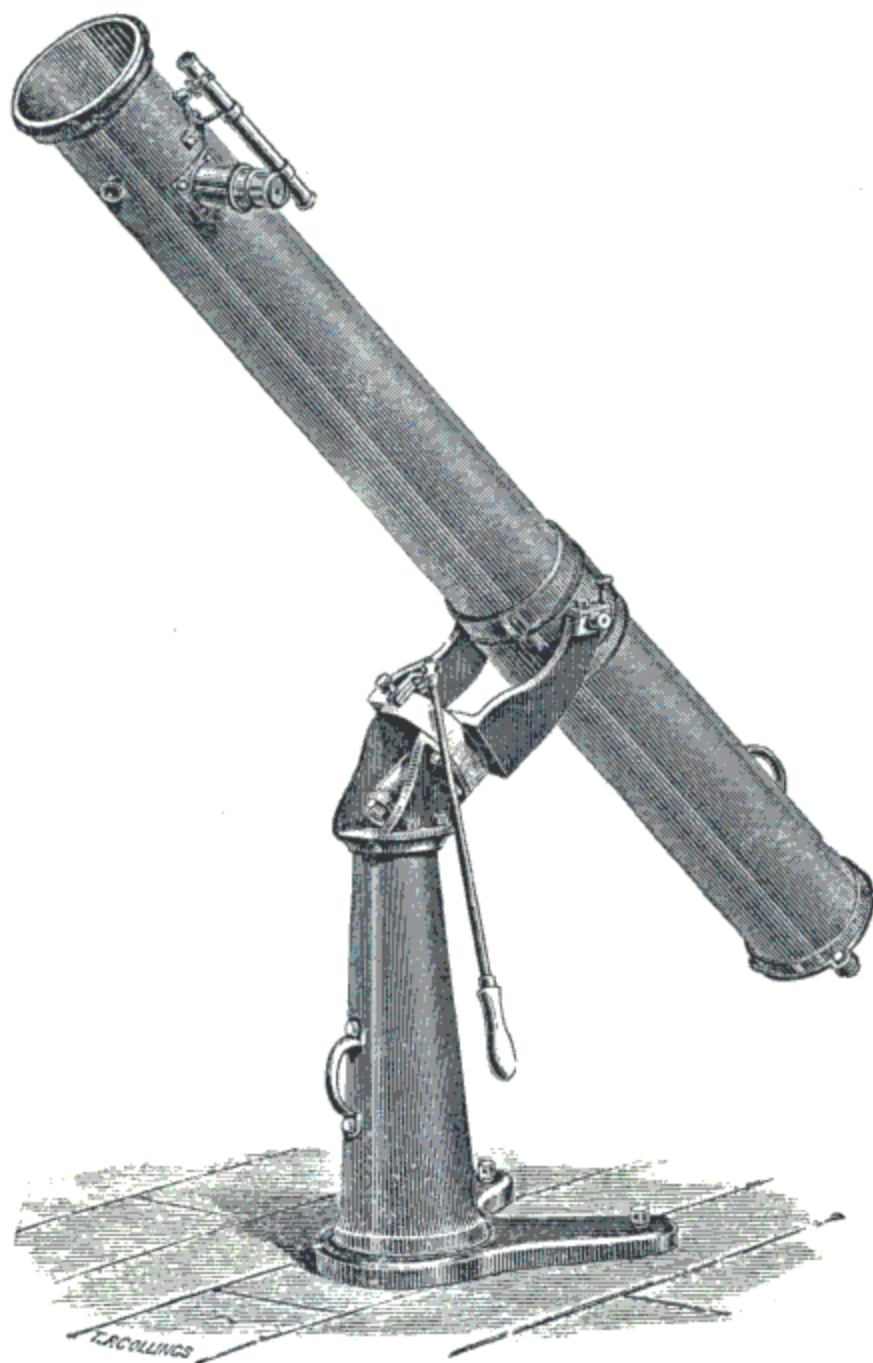
Mr. G. CALVER.



(FIG. 1.)



(FIG. 2.)



(FIG. 3.)

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TRANSCRIBER'S NOTE:

The constellation Boötis also appears in the original spelt as Bootis and Böötis these have all been corrected to Boötis without being noted below. Also variations in hyphenation remain unchanged.

The following typos are corrected in this e-text and are shown within the text as replaced by.

	Original	Replaced by
Page 4	Many of the speculas	Many of the specula
Page 7	on the pitch	or the pitch
Page 24	and proceed	and proceed.
Page 28	kept is glass	kept in glass
Page 31	See p. 27, Fig. 4	See p. 20, Fig. 4
Page 39 (appendix)	(prices for 8-1/2 then 6-1/2)	(prices for 6-1/2 then 8-1/2)
Page 14	Rev. W. P. MATTHEWS	REV. W. P. MATTHEWS
Page 24	(a 10 in. Alt Azimuth.	(a 10 in. Alt Azimuth).
Page 32	Rev. E. A. FISHBOURNE, M.A.	REV. E. A. FISHBOURNE, M.A.

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